

2.1 Introduction

In the study of the mechanics of particles, the fundamental kinematical variable that is used is the *position* of the body, and its two time derivatives, the *velocity* and the *acceleration*. The interaction of a given body with other bodies is quantified in terms of the *forces* that these other bodies exert on the first body. The effect that these forces have on the body is governed by Newton's law of motion, which states that the sum of the forces acting on a body is equal to the mass of the body times its acceleration. The condition for a body to be in equilibrium is that the sum of the external forces and moments acting on it must vanish.

These basic mechanical concepts such as position and force, as well as Newton's law of motion, also apply to extended, deformable bodies such as rock masses. However, these concepts must be altered somewhat, for various reasons. First, the fact that the force applied to a rock will, in general, vary from point to point, and will be distributed over the body must be taken into account. The idealization that forces act at localized points, which is typically used in the mechanics of particles, is not sufficiently general to apply to all problems encountered in rock mechanics. Hence, it is necessary to introduce the concept of *traction*, which is a force per unit area. As the traction generally varies with the orientation of the surface on which it acts, it is most conveniently represented with the aid of an entity known as the *stress tensor*.

Another fundamental difference between the mechanics of particles and deformable bodies such as rocks is that different parts of the rock may undergo different amounts of displacement. In general, it is the relative displacement of neighboring particles, rather than the absolute displacement of a particular particle, that can be equating in some way to the applied tractions. This can be seen from the fact that a rock sample can be moved *as a rigid body* from one location to another, after which the external forces acting on the rock can remain unaltered. Clearly, therefore, the displacement itself cannot be directly related to the applied loads. This relative displacement of nearby elements of the rock is quantified by an entity known as the *strain*.

The stress tensor is a symmetric second-order tensor, and many important properties of stress follow directly from those of second-order tensors. In the event that the relative displacements of all parts of the rock are small, the strain can also be represented by a second-order tensor called the *infinitesimal strain*

tensor. A consequence of this fact is that much of the general theory of stresses applies also to the analysis of strains. The general theory of stress and strain is the topic of this chapter. Both of these theories can be developed without any reference to the specific properties of the material under consideration (i.e., the constitutive relationship between the stress and strain tensors). Hence, the discussion given in this chapter parallels, to a great extent, that which is given in many texts on elasticity, or solid mechanics in general. Among the many classic texts on elasticity that include detailed discussion of the material presented in this chapter are Love (1927), Sokolnikoff (1956), Filonenko-Borodich (1965), and Timoshenko and Goodier (1970). The chapter ends with a brief introduction to the theory of finite strains.

2.2 Definition of traction and stress

Consider a rock mass that is subject to some arbitrary set of loads. At any given point within this rock, we can imagine a plane slicing through the rock at some angle. Such a plane may in fact form an external boundary of the rock mass, or may represent a fictitious plane that is entirely internal to the rock. Figure 2.1a shows such a plane, along with a fixed (x, y) coordinate system. In particular, consider an element of that plane that has area A . Most aspects of the theory of stress (and strain) can be developed within a two-dimensional context, and extensions to three dimensions are in most cases straightforward. As most figures are easier to draw, and to interpret, in two dimensions than in three, much of the following discussion will be given first in two-dimensional form.

The plane shown in Fig. 2.1a can be uniquely identified by the unit vector that is perpendicular to its surface. The vector $\mathbf{n} = (n_x, n_y)$ is the *outward unit normal vector* to this plane: it has *unit* length, is *normal* to the plane, and points in the direction *away* from the rock mass. The components of this vector $\mathbf{n}$ are the direction cosines that the outward unit normal vector makes with the two coordinate axes. For example, a plane that is perpendicular to the x -axis would have $\mathbf{n} = (1, 0)$. As the length of any unit normal vector is unity, the Pythagorean theorem implies that $(n_x)^2 + (n_y)^2 = 1$. The unit normal vectors in the directions of the coordinate axes are often denoted by $\mathbf{e}_x = (1, 0)$ and $\mathbf{e}_y = (0, 1)$. The identification of a plane by its outward unit normal vector is employed frequently

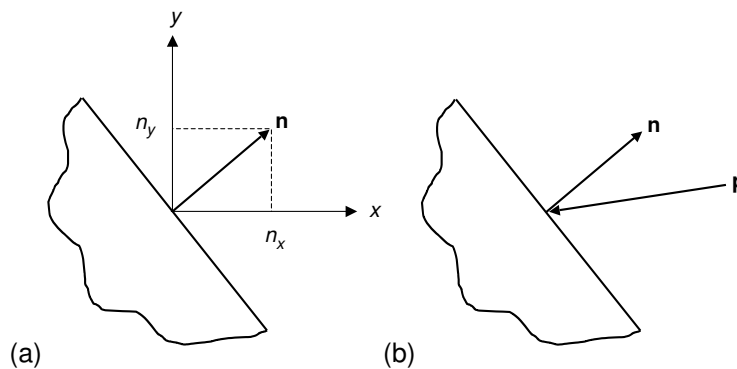


Fig. 2.1 Normal vector $\mathbf{n}$ and traction vector $\mathbf{p}$ acting on a surface.

in rock mechanics. It is important to remember that the vector $\mathbf{n}$ is perpendicular to the plane in question; it does not lie within that plane.

The action that the rock adjacent to the plane exerts on the rock that is “interior” to the plane can be represented by a resultant force $\mathbf{F}$, which, like all forces, is a vector. The traction vector $\mathbf{p}$ is defined as the ratio of the resultant force $\mathbf{F}$ to the surface area A :

$$\mathbf{p}(\text{averaged over the area}) = \frac{1}{A}\mathbf{F}. \quad (2.1)$$

In order to define the traction that acts over a specific “point” in the rock, the area is now allowed to shrink down to a point, so that the magnitude A goes to zero. Following the convention often used in applied mathematics, the smallness of the area is indicated by the notation “ dA ,” where the “ d ” denotes “differential,” and likewise for the resultant force $\mathbf{F}$. As the area shrinks down to a point, the traction at that point can then be defined by (Fig. 2.1b)

$$\mathbf{p}(\mathbf{x}; \mathbf{n}) = \lim_{dA \rightarrow 0} \frac{1}{dA} d\mathbf{F}. \quad (2.2)$$

The notation $\mathbf{p}(\mathbf{x}; \mathbf{n})$ denotes the traction vector at the point $\mathbf{x} \equiv (x, y, z)$, on a plane whose outward unit normal vector is $\mathbf{n}$. In the following discussion, when the point $\mathbf{x}$ under consideration is either clear from the context, or immaterial to the particular discussion, the dependence of $\mathbf{p}$ on $\mathbf{x}$ will be suppressed in the notation.

At this point, it is necessary to introduce a sign convention that is inconsistent with the one used in most areas of mechanics, but which is nearly universal in the study of rocks and soils. The Cartesian component of the traction $\mathbf{p}$ in any given direction $\mathbf{r}$ is considered to be a positive number if the inner product (dot product) of $\mathbf{p}$ and a unit vector in the $\mathbf{r}$ direction is *negative*. One way to interpret this convention is that the traction is based on $-\mathbf{F}$, rather than $\mathbf{F}$. The reason for utilizing this particular sign convention will become clear after the stresses are introduced.

It is apparent from the definition given in (2.2) that the traction is a vector, and therefore has two components in a two-dimensional system, and three components in a three-dimensional system. In general, this vector may vary from point to point, and is therefore a function of the location of the point in question. However, at any given point, the traction will also, in general, be different on different planes that pass through that point. In other words, the traction will also be a function of $\mathbf{n}$, the outward unit normal vector. The fact that $\mathbf{p}$ is a function of two vectors, the position vector $\mathbf{x}$ and the outward unit normal vector $\mathbf{n}$, is awkward. This difficulty is eliminated by appealing to the concept of *stress*, which was introduced in 1823 by the French civil engineer and mathematician Cauchy. The stress concept allows all possible traction vectors at a point to be represented by a single mathematical entity that does not explicitly depend on the unit normal of any particular plane. The price paid for this simplification, so to speak, is that the stress is not a vector, but rather a second-order tensor, which is a somewhat more complicated, and less familiar, mathematical object than is a vector.

Although there are an infinite number of different traction vectors at a point, corresponding to the infinity of possible planes passing through that point, all possible traction vectors can be found from knowledge of the traction vector on two mutually orthogonal planes (or three mutually orthogonal planes in three dimensions). To derive the relationship for the traction on an arbitrary plane, it is instructive to follow the arguments originally put forward by Cauchy. Consider a thin penny-shaped slab of rock having thickness h , and radius r (Fig. 2.2a). The outward unit normal vector on the right face of this slab is denoted by $\mathbf{n}$; the outward unit normal vector of the left face of the slab is therefore $-\mathbf{n}$. The total force acting on the face with outward unit normal vector $\mathbf{n}$ is equal to $\pi r^2 \mathbf{p}(\mathbf{n})$, whereas the total force acting on the opposing face is $\pi r^2 \mathbf{p}(-\mathbf{n})$. The total force acting on the outer rim of this penny-shaped slab will be given by an integral of the traction over the outer area, and will be proportional to $2\pi rh$, which is the surface area of the outer rim. Performing a force balance on this slab of rock yields

$$\pi r^2 \mathbf{p}(\mathbf{n}) + \pi r^2 \mathbf{p}(-\mathbf{n}) + 2\pi rh \mathbf{t} = 0, \quad (2.3)$$

where $\mathbf{t}$ is the mean traction over the outer rim. If the thickness h of this slab is allowed to vanish, this third term will become negligible, and the condition for equilibrium becomes

$$\mathbf{p}(-\mathbf{n}) = -\mathbf{p}(\mathbf{n}). \quad (2.4)$$

Equation (2.4), known as *Cauchy's first law*, essentially embodies a version of Newton's third law: if the material to the left of a given plane exerts a traction $\mathbf{p}$ on the material on the right, then the material on the right will exert a traction $-\mathbf{p}$ on the material to the left.

Now, consider a triangular slab of rock, as in Fig. 2.2b, with a uniform thickness w in the third (z) direction. Two faces of this slab have outward unit normal vectors that coincide with the negative x and y coordinate directions, respectively, whereas the third face has an outward unit normal vector of $\mathbf{n} = (n_x, n_y)$. The length of the face with outward unit normal vector $\mathbf{n}$ is taken to be h . The length of the face that has outward unit normal vector $\mathbf{n} = -\mathbf{e}_x = (-1, 0)$ is equal

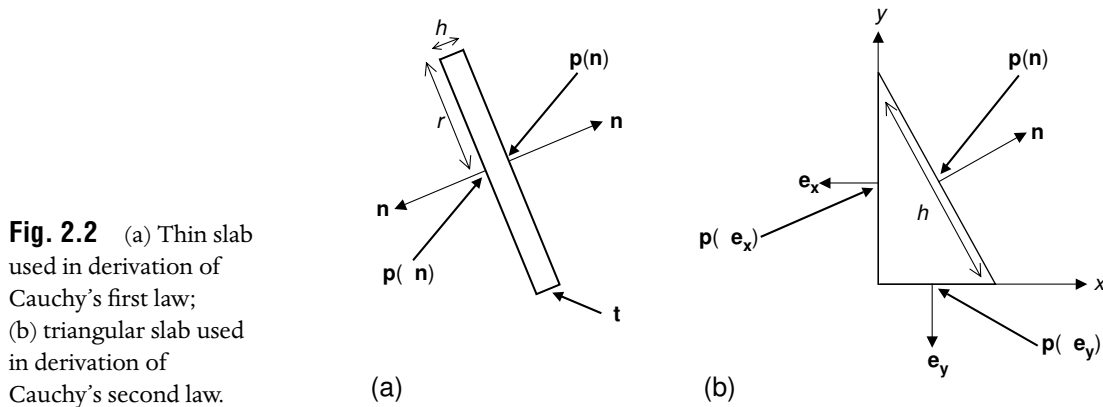


Fig. 2.2 (a) Thin slab used in derivation of Cauchy's first law; (b) triangular slab used in derivation of Cauchy's second law.

to hn_x , and so its area is hwn_x . The traction vector on this face is denoted by $\mathbf{p} = (-\mathbf{e}_x)$, and so the total force acting on this face is $hwn_x\mathbf{p}(-\mathbf{e}_x)$. Similar considerations show that the total force acting on the face with outward unit normal vector $-\mathbf{e}_y$ will be $hwn_y\mathbf{p}(-\mathbf{e}_y)$. Hence, a force balance on this slab leads to

$$hwn_x\mathbf{p}(-\mathbf{e}_x) + hwn_y\mathbf{p}(-\mathbf{e}_y) + hw\mathbf{p}(\mathbf{n}) = 0. \quad (2.5)$$

Canceling out the common terms, and utilizing Cauchy's first law, (2.4), leads to *Cauchy's second law*:

$$\mathbf{p}(\mathbf{n}) = n_x\mathbf{p}(\mathbf{e}_x) + n_y\mathbf{p}(\mathbf{e}_y). \quad (2.6)$$

This result would remain unchanged if we consider the more general case in which a distributed *body force* acts on the tetrahedral-shaped element as in Fig. 2.2b. Whereas surface forces act over the outer surface of an element of rock, body forces act over the entire volume of the rock. The most obvious and common body force encountered in rock mechanics is that due to gravity, which has a magnitude of ρg (per unit volume), and is directed in the downward vertical direction. However, as will be shown in Chapter 7, gradients in temperature and pore fluid pressure also give rise to phenomena which have the same effect as distributed body forces. If there were a body-force density $\mathbf{b}$ per unit volume of rock, a total body force of $(1/2)h^2wn_xn_y\mathbf{b}$ would have to be added to the force balance in (2.5). If we divide through by h , and then let the size of the element shrink to zero (i.e., $h \rightarrow 0$), the body force term would drop out and $\mathbf{b}$ would not appear in the final result (2.6).

It is now convenient to recall that each traction is a vector, and therefore (in two dimensions) will have two components, one in each of the coordinate directions. The components of a traction vector such as $\mathbf{p}(\mathbf{e}_x)$ are denoted using two indices – the first to indicate the direction of the outward unit normal vector and the second to indicate the component of the traction vector:

$$\mathbf{p}(\mathbf{e}_x) = \begin{bmatrix} \tau_{xx} \\ \tau_{xy} \end{bmatrix} = [\tau_{xx} \quad \tau_{xy}]^T, \quad (2.7)$$

where we adhere to the algebraic convention that a vector is written as a column, and is therefore equivalent to the *transpose* of a row vector. Equation (2.6) can therefore be written in matrix form as

$$\mathbf{p}(\mathbf{n}) = n_x \begin{bmatrix} \tau_{xx} \\ \tau_{xy} \end{bmatrix} + n_y \begin{bmatrix} \tau_{yx} \\ \tau_{yy} \end{bmatrix} = \begin{bmatrix} \tau_{xx} & \tau_{yx} \\ \tau_{xy} & \tau_{yy} \end{bmatrix} \begin{bmatrix} n_x \\ n_y \end{bmatrix}. \quad (2.8)$$

In the first expression on the right in (2.8), n_x and n_y are treated as scalars that multiply the two traction vectors; in the second expression, the formalism of matrix multiplication is used. As the two components of the vector $\mathbf{p}(\mathbf{n})$ are $p_x(\mathbf{n})$ and $p_y(\mathbf{n})$, (2.8) can be written in component form as

$$p_x(\mathbf{n}) = \tau_{xx}n_x + \tau_{yx}n_y, \quad (2.9)$$

$$p_y(\mathbf{n}) = \tau_{xy}n_x + \tau_{yy}n_y. \quad (2.10)$$

If we use the standard matrix algebra convention that the first subscript of a matrix component denotes the row, and the second subscript denotes the column, the matrix appearing in (2.8) is seen to actually be the *transpose* of the stress matrix, in which case we can rewrite (2.8) as

$$\mathbf{p}(\mathbf{n}) = \begin{bmatrix} \tau_{xx} & \tau_{xy} \\ \tau_{yx} & \tau_{yy} \end{bmatrix}^T \begin{bmatrix} n_x \\ n_y \end{bmatrix}, \quad (2.11)$$

where the matrix that appears in (2.11), without the transpose operator, is the stress matrix, $\boldsymbol{\tau}$. Equation (2.11) can be written in direct matrix notation as

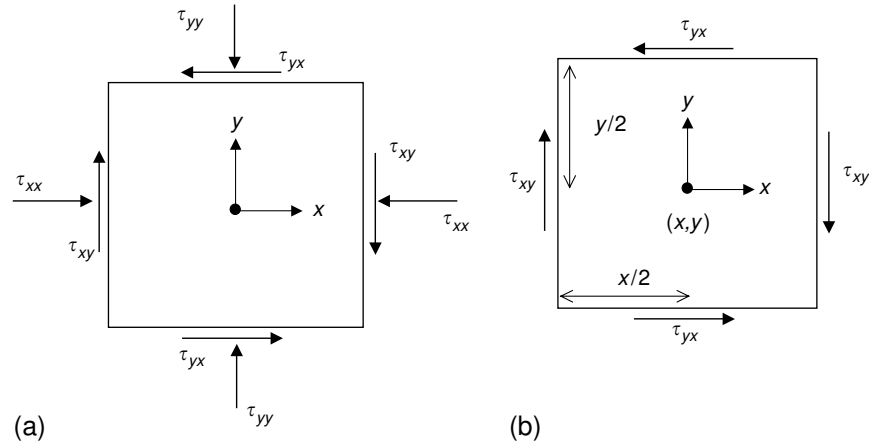
$$\mathbf{p} = \boldsymbol{\tau}^T \mathbf{n}, \quad (2.12)$$

where $\mathbf{n}$ is a unit normal vector, $\mathbf{p}$ is the traction vector on the plane whose outward unit normal vector is $\mathbf{n}$, and $\boldsymbol{\tau}$ is the stress matrix, or *stress tensor*. In two dimensions the stress tensor has four components; in three dimensions it has nine. Equation (2.12) gives the traction on an arbitrarily oriented plane in terms of the stress matrix, relative to some fixed orthogonal coordinate system, and the direction cosines between the outward unit normal vector to the plane and the two coordinate axes. Note that a tensor can be written as a matrix, which is merely a rectangular array of numbers. However, a tensor has specific mathematical properties that are not necessarily shared by an arbitrary matrix-like collection of numbers. These properties relate to the manner in which the components of a tensor transform when the coordinate system is changed; these transformation laws are discussed in more detail in §2.3. The rows of the matrix that represents $\boldsymbol{\tau}$ are the traction vectors along faces whose outward unit normal vectors lie along the coordinate axes. In other words, the first row of $\boldsymbol{\tau}$ is $\mathbf{p}(\mathbf{e}_x)$, the second row is $\mathbf{p}(\mathbf{e}_y)$, etc.

The physical significance of the stress tensor is traditionally illustrated by the schematic diagram shown in Fig. 2.3a. Consider a two-dimensional square element of rock, whose faces are each perpendicular to one of the two coordinate axes. The traction vector that acts on the face whose outward unit normal vector is in the x direction has components (τ_{xx}, τ_{xy}) . Each of these two components can be considered as a vector in its own right; they are indicated in Fig. 2.3a as arrows whose lines of action pass through the center of the face whose outward unit normal vector is $\mathbf{e}_x$. As the traction components are considered positive if they are oriented in the directions opposite to the outward unit normal vector, we see that the traction τ_{xx} is a positive number if it is *compressive*. Compressive stresses are much more common in rock mechanics than are tensile stresses. For example, the stresses in a rock mass that are due to the weight of the overlying rock are compressive. In most other areas of mechanics, tensile stresses are considered positive, and compressive stresses are reckoned to be negative. The opposite sign convention is traditionally used in rock (and soil) mechanics in order to avoid the frequent occurrence of negative signs in calculations involving stresses.

Many different notations have been used to denote the components of the stress tensor. We will mainly adhere to the notation introduced above, which has

Fig. 2.3 (a) Stress components acting on a small square element. (b) Balance of angular momentum on this element shows that the stress tensor must be symmetric.



been used, for example, by Sokolnikoff (1956). Some authors use σ instead of τ as the basic symbol, but utilize the same subscripting convention. Many rock and soil mechanics treatments, including earlier editions of this book, denote shear stresses by τ_{xy} , etc., but denote normal stresses by, for example, σ_x rather than τ_{xx} . This notation, which has also been used by Timoshenko and Goodier (1970), has the advantage of clearly indicating the distinction between normal and shear components of the stress, which have very different physical effects, particularly when acting on fracture planes or other planes of weakness (Chapter 3). However, the $\{\sigma, \tau\}$ notation does not reflect the fact that the normal and shear components of the stress are both components of a single mathematical object known as the stress tensor. Furthermore, many of the equations in rock mechanics take on a simpler and more symmetric form if written in terms of a notation in which all stress components are written using the same symbol. However, a version of the Timoshenko and Goodier notation will occasionally be used in this book when discussing the traction acting on a specific plane. In such cases, for reasons of simplicity (so as to avoid the need for subscripts), it will be convenient to denote the normal stress by σ , and the shear stress by τ . Many classic texts on elasticity, such as Love (1927) and Filonenko-Borodich (1965), utilize the notation introduced by Kirchhoff in which τ_{xy} is denoted by X_y , etc. Green and Zerna (1954) use the notation suggested by Todhunter and Pearson (1886), in which τ_{xy} is denoted by $\hat{x}\hat{y}$, etc.

Equation (2.12) is usually written without the transpose sign, although strictly speaking the transpose is needed. The reason that it is allowable to write $\mathbf{p} = \boldsymbol{\tau}\mathbf{n}$ in place of $\mathbf{p} = \boldsymbol{\tau}^T\mathbf{n}$ is that the stress matrix is in fact always *symmetric*, so that $\tau_{xy} = \tau_{yx}$, in which case $\boldsymbol{\tau} = \boldsymbol{\tau}^T$. This property of the stress tensor is of great importance, if for no other reason than that it reduces the number of stress components that must be measured or calculated from four to three in two dimensions, and from nine to six in three dimensions. The symmetry of the stress tensor can be proven by appealing to the law of conservation of angular momentum. For simplicity, consider a rock subject to a state of stress that does not vary from point to point. If we draw a free-body diagram for a small

rectangular element of rock, centered at point (x, y) , the traction components acting on the four faces are shown in Fig. 2.3b. The length of the element is Δx in the x direction, Δy in the y direction, and Δz in the z direction (into the page). In order for this element of rock to be in equilibrium, the sum of all the moments about any point, such as (x, y) , must be zero. Consider first the tractions that act on the right face of the element. The force vector represented by this traction is found by multiplying the traction by the area of that face, which is $\Delta y \Delta z$. The x -component of this force is therefore $\tau_{xx} \Delta y \Delta z$. However, the resultant of this force acts through the point (x, y) , and therefore contributes no moment about that point. The y -component of this traction is τ_{xy} , and the net force associated with it is $\tau_{xy} \Delta y \Delta z$. The moment arm of this force is $\Delta x/2$, so that the total clockwise moment about the z -axis, through the point (x, y) , is $\tau_{xy} \Delta x \Delta y \Delta z/2$. Adding up the four moments that are contributed by the four shear stresses yields

$$\tau_{xy} \Delta x \Delta y \Delta z/2 + \tau_{yx} \Delta y \Delta x \Delta z/2 - \tau_{yx} \Delta x \Delta y \Delta z/2 - \tau_{xy} \Delta y \Delta x \Delta z/2 = 0. \quad (2.13)$$

Canceling out the terms $\Delta x \Delta y \Delta z/2$ leads to the result

$$\tau_{xy} = \tau_{yx}. \quad (2.14)$$

In three dimensions, a similar analysis leads to the relations $\tau_{xz} = \tau_{zx}$ and $\tau_{yz} = \tau_{zy}$. This result should be interpreted as stating that at any specific point (x, y, z) , the stress component $\tau_{xy}(x, y, z)$ is equal in magnitude and sign to the stress component $\tau_{yx}(x, y, z)$. There is in general no reason for the conjugate shear stresses *at different points* to be equal to each other.

Although the derivation presented above assumes that the stresses do not vary from point to point, and that the element of rock is in static equilibrium, the result is actually completely general. The reason for this is related to the fact that the result applies at each infinitesimal “point” in the rock. If we had accounted for the variations of the stress components with position, these terms would contribute moments that are of higher order in Δx and Δy . Dividing through the moment balance equation by $\Delta x \Delta y \Delta z$, and then taking the limit as Δx and Δy go to zero, would cause these additional terms to drop out, leading to (2.14). The same would occur if we considered the more general situation in which the element were not in static equilibrium, but rather was rotating. In this situation, the sum of the moments would be equal to the moment of inertia of the element about the z -axis through the point (x, y) , which is $\rho \Delta x \Delta y \Delta z [(\Delta x)^2 + (\Delta y)^2]/12$, where ρ is the density of the rock, multiplied by the angular acceleration, $\dot{\omega}$. Hence, the generalization of (2.14) would be

$$\tau_{xy} \Delta x \Delta y \Delta z - \tau_{yx} \Delta x \Delta y \Delta z = \rho \Delta x \Delta y \Delta z [(\Delta x)^2 + (\Delta y)^2] \dot{\omega}/12. \quad (2.15)$$

Dividing through by $\Delta x \Delta y \Delta z$, and then taking the limit as the element shrinks down to the point (x, y) , leads again to (2.14).

The symmetry of the stress tensor is therefore a general result. However, it is worth bearing in mind that although τ_{xy} and τ_{yx} are numerically equal, they are in fact physically distinct stress components, and act on different faces of an element

of rock. Although the identification of τ_{xy} with τ_{yx} is eventually made when solving the elasticity equations, it is usually preferable to maintain a distinction between τ_{xy} and τ_{yx} when writing out equations, or drawing schematic diagrams such as Fig. 2.3a. This helps to preserve as much symmetry as possible in the structure of the equations.

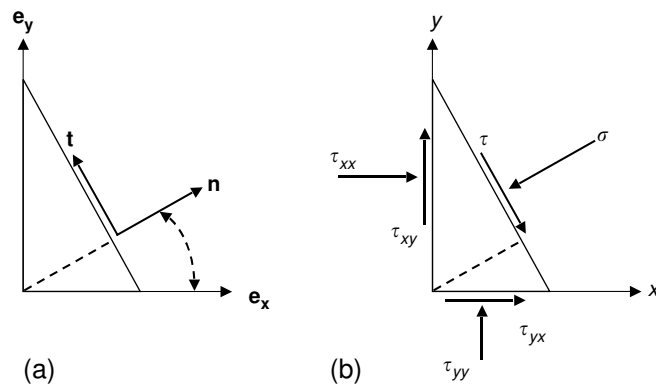
The symmetry of the stress tensor followed from the principle of conservation of angular momentum. The principle of conservation of *linear* momentum leads to three further equations that must be satisfied by the stresses. These equations, which are known as the *equations of stress equilibrium* and are derived in §5.5, control the rate at which the stresses vary in space. However, much useful information about the stress tensor can be derived prior to considering the implications of the equations of stress equilibrium. Of particular importance are the laws that govern the manner in which the stress components vary as the coordinate system is rotated. These laws are derived and discussed in §2.3 and §2.5.

2.3 Analysis of stress in two dimensions

Discussions of stress are algebraically simpler in two dimensions than in three. In most instances, no generality is lost by considering the two-dimensional case, as the extension to three dimensions is usually straightforward. Furthermore, many problems in rock mechanics are essentially two dimensional, in the sense that the stresses do not vary along one Cartesian coordinate. The most common examples of such problems are stresses around boreholes, or around long tunnels. Many other problems are idealized as being two dimensional so as to take advantage of the relative ease of solving two-dimensional elasticity problems as compared to three-dimensional problems. Hence, it is worthwhile to study the properties of two-dimensional stress tensors. Various properties of two-dimensional stress tensors will be examined in this section; their three-dimensional analogues will be discussed in §2.5.

In order to derive the laws that govern the transformation of stress components under a rotation of the coordinate system, we again consider a small triangular element of rock, as in Fig. 2.4. The outward unit normal vector to

Fig. 2.4 Small triangular slab of rock used to derive the stress transformation equations.



the slanted face of the element is $\mathbf{n} = (n_x, n_y)$. We can construct another unit vector $\mathbf{t}$, perpendicular to $\mathbf{n}$, which lies along this face. Being of unit length, the components of $\mathbf{t}$ must satisfy the condition $\mathbf{t} \cdot \mathbf{t} = (t_x)^2 + (t_y)^2 = 1$. The orthogonality of $\mathbf{t}$ and $\mathbf{n}$ implies that $\mathbf{t} \cdot \mathbf{n} = t_x n_x + t_y n_y = 0$, which shows that $\mathbf{t} = \pm(n_y, -n_x)$. Finally, if we require that the pair of vectors $\{\mathbf{n}, \mathbf{t}\}$ have the same orientation relative to each other as do the pair $\{\mathbf{e}_x, \mathbf{e}_y\}$, the minus sign must be used, in which case $\mathbf{t} = (-n_y, n_x)$. This pair of vectors can be thought of as forming a new Cartesian coordinate system that is rotated from the original (x, y) system by a counterclockwise angle of $\theta = \arccos(n_x)$. According to (2.9) and (2.10), the components of the traction vector $\mathbf{p}(\mathbf{n})$, expressed in terms of the (x, y) coordinate system, are given by

$$p_x = \tau_{xx}n_x + \tau_{yx}n_y, \quad (2.16)$$

$$p_y = \tau_{xy}n_x + \tau_{yy}n_y. \quad (2.17)$$

In order to find the components of $\mathbf{p}$ relative to the $\{\mathbf{n}, \mathbf{t}\}$ coordinate system, we take the inner products of $\mathbf{p}$ with respect to $\mathbf{n}$ and $\mathbf{t}$, in turn. For example,

$$p_n = \mathbf{p} \cdot \mathbf{n} = p_x n_x + p_y n_y = \tau_{xx}n_x^2 + \tau_{yx}n_y n_x + \tau_{xy}n_x n_y + \tau_{yy}n_y^2. \quad (2.18)$$

Utilization of the symmetry property $\tau_{yx} = \tau_{xy}$ allows this to be written as

$$p_n = \tau_{xx}n_x^2 + 2\tau_{xy}n_x n_y + \tau_{yy}n_y^2. \quad (2.19)$$

Similarly, the tangential component of the traction vector on this face, which is given by $p_t = \mathbf{p} \cdot \mathbf{t}$, can be expressed as

$$p_t = (\tau_{yy} - \tau_{xx})n_x n_y + \tau_{xy}(n_x^2 - n_y^2). \quad (2.20)$$

The two unit vectors $\{\mathbf{n}, \mathbf{t}\}$ can be thought of as defining a new coordinate system that is rotated by a counterclockwise angle θ from the old coordinate system. This interpretation is facilitated by denoting these two new unit vectors by $\{\mathbf{e}_{x'}, \mathbf{e}_{y'}\}$. Equations (2.19) and (2.20) are therefore seen to give the components of the traction vector on the plane whose outward unit vector is $\mathbf{e}_{x'}$, that is,

$$p_{x'}(\mathbf{e}_{x'}) \equiv \tau_{x'x'} = \tau_{xx}n_x^2 + 2\tau_{xy}n_x n_y + \tau_{yy}n_y^2, \quad (2.21)$$

$$p_{y'}(\mathbf{e}_{x'}) \equiv \tau_{x'y'} = (\tau_{yy} - \tau_{xx})n_x n_y + \tau_{xy}(n_x^2 - n_y^2), \quad (2.22)$$

where, for clarity, we reemphasize that these components pertain to the traction on the plane with outward unit normal vector $\mathbf{e}_{x'}$. According to the discussion given in §2.2, these components can also be interpreted as the components of the stress tensor in the (x', y') coordinate system. Specifically, $p_{x'}(\mathbf{e}_{x'}) = \tau_{x'x'}$, and $p_{y'}(\mathbf{e}_{x'}) = \tau_{x'y'}$. The traction vector on the plane whose outward unit normal vector is $\mathbf{e}_{y'}$ can be found by a similar analysis; the results are

$$p_{y'}(\mathbf{e}_{y'}) \equiv \tau_{y'y'} = \tau_{xx}n_y^2 - 2\tau_{xy}n_x n_y + \tau_{yy}n_x^2, \quad (2.23)$$

$$p_{x'}(\mathbf{e}_{y'}) \equiv \tau_{y'x'} = (\tau_{yy} - \tau_{xx})n_x n_y + \tau_{xy}(n_x^2 - n_y^2). \quad (2.24)$$

Note that $p_{x'}(\mathbf{e}_{y'}) = \tau_{y'x'} = p_{y'}(\mathbf{e}_{x'}) = \tau_{x'y'}$, as must necessarily be the case, due to the general property of symmetry of the stress tensor.

Another common notation used for the stress transformation equations in two dimensions can be obtained by recalling that the primed coordinate system is derived from the unprimed system by rotation through a counterclockwise angle of $\theta = \arccos(n_x)$. Furthermore, the components (n_x, n_y) of the unit normal vector $\mathbf{n}$ can be written as $(\cos \theta, \sin \theta)$. In terms of the angle of rotation, the stresses in the rotated coordinate system are

$$\tau_{x'x'} = \tau_{xx} \cos^2 \theta + 2\tau_{xy} \sin \theta \cos \theta + \tau_{yy} \sin^2 \theta, \quad (2.25)$$

$$\tau_{y'y'} = \tau_{xx} \sin^2 \theta - 2\tau_{xy} \sin \theta \cos \theta + \tau_{yy} \cos^2 \theta, \quad (2.26)$$

$$\tau_{x'y'} = (\tau_{yy} - \tau_{xx}) \sin \theta \cos \theta + \tau_{xy}(\cos^2 \theta - \sin^2 \theta). \quad (2.27)$$

This rotation operation can be represented by the rotation matrix $\mathbf{L}$, which has the defining properties that $\mathbf{L}^T \mathbf{e}_x = \mathbf{e}_{x'}$, and $\mathbf{L}^T \mathbf{e}_y = \mathbf{e}_{y'}$. In component form, relative to the (x, y) coordinate system, the two primed unit vectors are given by $\mathbf{e}_{x'} = (\cos \theta, \sin \theta)$ and $\mathbf{e}_{y'} = (-\sin \theta, \cos \theta)$. These two vectors therefore form the two columns of the matrix $\mathbf{L}^T$ (Lang, 1971, p. 120), which is to say they form the rows of $\mathbf{L}$, that is,

$$\mathbf{L} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}. \quad (2.28)$$

Using this rotation matrix, the transformation equations (2.25)–(2.27) can be written in the following matrix form:

$$\begin{bmatrix} \tau_{x'x'} & \tau_{x'y'} \\ \tau_{y'x'} & \tau_{y'y'} \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \tau_{xx} & \tau_{xy} \\ \tau_{yx} & \tau_{yy} \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}, \quad (2.29)$$

which can also be expressed in direct matrix notation as

$$\boldsymbol{\tau}' = \mathbf{L} \boldsymbol{\tau} \mathbf{L}^T. \quad (2.30)$$

The fact that the stresses transform according to (2.30) when the coordinate system is rotated is the defining property that makes the stress a *second-order tensor*. We note also that, using this direct matrix notation, the traction vector transforms according to $\mathbf{p}' = \mathbf{L} \mathbf{p}$. The appearance of *one* rotation matrix in this transformation law is the reason that vectors are referred to as *first-order tensors*.

The form of the stress transformation law given in (2.29) or (2.30) is convenient when considering a rotation of the coordinate system. However, from a more physically based viewpoint, it is pertinent to focus attention on the tractions that act on a given plane, such as the one shown in Fig. 2.4. The same equations are used in both situations, but their interpretation is slightly different. When focusing on a specific plane with unit normal vector $\mathbf{n}$, it is convenient to simplify the equations by utilizing the trigonometric identities $\cos^2 \theta - \sin^2 \theta = \cos 2\theta$, and $2 \sin \theta \cos \theta = \sin 2\theta$. As long as attention is focused on a given plane, no confusion should arise if the normal stress acting on this plane is denoted by σ , and the shear stress is denoted by τ . After some algebraic manipulation, we

arrive at the following equations for the normal and shear stresses acting on a plane whose outward unit normal vector is rotated counterclockwise from the x direction by an angle θ :

$$\sigma = \frac{1}{2}(\tau_{xx} + \tau_{yy}) + \frac{1}{2}(\tau_{xx} - \tau_{yy}) \cos 2\theta + \tau_{xy} \sin 2\theta, \quad (2.31)$$

$$\tau = \frac{1}{2}(\tau_{yy} - \tau_{xx}) \sin 2\theta + \tau_{xy} \cos 2\theta. \quad (2.32)$$

The variation of σ and τ with the angle of rotation is illustrated in Fig. 2.5, for the case where $\{\tau_{xx} = 4, \tau_{yy} = 2, \tau_{xy} = 1\}$.

An interesting question to pose is whether or not there are planes on which the shear stress vanishes, and where the stress therefore has purely a normal component. The answer follows directly from setting $\tau = 0$ in (2.32), and solving for

$$\tan 2\theta = \frac{2\tau_{xy}}{\tau_{xx} - \tau_{yy}}. \quad (2.33)$$

If $\tau_{xy} = 0$, then the plane with $\mathbf{n} = \mathbf{e}_x$ is already a shear-free plane, and (2.33) gives the result $\theta = 0$. In general, however, whatever the values of $\{\tau_{xy}, \tau_{xx}, \tau_{yy}\}$, there will always be two roots of (2.33) in the range $0 \leq 2\theta < 2\pi$, and these roots will differ by π . Hence, there will be two values of θ that satisfy (2.33), differing by $\pi/2$, and lying in the range $0 \leq \theta < \pi$; this situation will be discussed in more detail below. For now, note that if θ is defined by (2.33), it follows from elementary trigonometry that

$$\sin 2\theta = \pm[1 + \cos^2 2\theta]^{-1/2} = \pm\tau_{xy}[\tau_{xy}^2 + \frac{1}{4}(\tau_{xx} - \tau_{yy})^2]^{-1/2}, \quad (2.34)$$

$$\cos 2\theta = \pm[1 + \tan^2 2\theta]^{-1/2} = \pm\frac{1}{2}(\tau_{xx} - \tau_{yy})[\tau_{xy}^2 + \frac{1}{4}(\tau_{xx} - \tau_{yy})^2]^{-1/2}, \quad (2.35)$$

in which case the normal stress is found from (2.31) to be given by

$$\sigma = \frac{1}{2}(\tau_{xx} + \tau_{yy}) \pm [\tau_{xy}^2 + \frac{1}{4}(\tau_{xx} - \tau_{yy})^2]^{-1/2}. \quad (2.36)$$

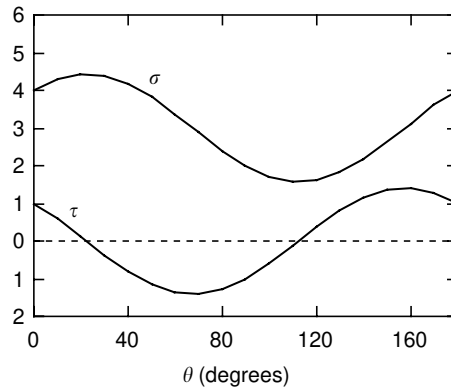


Fig. 2.5 Variation of normal and shear tractions with the angle θ (see Fig. 2.4a).

Equation (2.36) defines two normal stresses, σ_1 and σ_2 , that are known as the *principal normal stresses*, or simply the *principal stresses*. These stresses act on planes whose orientations relative to the (x, y) coordinate system are given by (2.33). It is customary to set $\sigma_1 \geq \sigma_2$, in which case the $+$ sign in (2.36) is associated with σ_1 , that is,

$$\sigma_1 = \frac{1}{2}(\tau_{xx} + \tau_{yy}) + [\tau_{xy}^2 + \frac{1}{4}(\tau_{xx} - \tau_{yy})^2]^{1/2}, \quad (2.37)$$

$$\sigma_2 = \frac{1}{2}(\tau_{xx} + \tau_{yy}) - [\tau_{xy}^2 + \frac{1}{4}(\tau_{xx} - \tau_{yy})^2]^{1/2}. \quad (2.38)$$

These two principal normal stresses not only have the distinction of acting on planes on which there is no shear, but are also the minimum and maximum normal stresses that act on *any* planes through the point in question. This can be proven by noting that

$$\frac{d\sigma}{d\theta} = -(\tau_{xx} - \tau_{yy}) \sin 2\theta + 2\tau_{xy} \cos 2\theta = -2\tau, \quad (2.39)$$

so that any plane on which τ vanishes is also a plane on which σ takes on a locally extreme value. This is apparent from Fig. 2.5, which also shows, for example, that the shear traction τ will take on its maximum and minimum values on two orthogonal planes whose normal vectors bisect the two directions of principal normal stress.

Although it is clear from (2.37) and (2.38) which of the two principal stresses is largest, the direction in which the major principal stress acts is not so clear, due to the fact that (2.33) has two physically distinct solutions, that differ by $\pi/2$. The correct choice for σ_1 is the angle that makes the normal stress a local maximum, rather than a local minimum. To determine the correct value we examine the second derivatives of σ with respect to θ . From (2.39),

$$\frac{d^2\sigma}{d\theta^2} = -2(\tau_{xx} - \tau_{yy}) \cos 2\theta - 4\tau_{xy} \sin 2\theta. \quad (2.40)$$

Using (2.40), along with (2.33), eventually leads to the following results (Chou and Pagano, 1992, p. 10):

$$\tau_{xx} > \tau_{yy} \quad \text{and} \quad \tau_{xy} > 0 \quad \Rightarrow \quad 0 < \theta_1 < 45^\circ, \quad (2.41)$$

$$\tau_{xx} < \tau_{yy} \quad \text{and} \quad \tau_{xy} > 0 \quad \Rightarrow \quad 45^\circ < \theta_1 < 90^\circ, \quad (2.42)$$

$$\tau_{xx} < \tau_{yy} \quad \text{and} \quad \tau_{xy} < 0 \quad \Rightarrow \quad 90^\circ < \theta_1 < 135^\circ, \quad (2.43)$$

$$\tau_{xx} > \tau_{yy} \quad \text{and} \quad \tau_{xy} < 0 \quad \Rightarrow \quad 135^\circ < \theta_1 < 180^\circ. \quad (2.44)$$

The principal stresses and principal directions can also be found by a different method, which can more readily be generalized to three dimensions. We start again by asking whether or not there are planes on which the traction vector is purely normal, with no shear component. On such planes, the traction vector will be parallel to the outward unit normal vector, and can therefore be expressed as $\mathbf{p} = \sigma \mathbf{n}$, where σ is some (as yet unknown) scalar. From (2.27) it is known that $\mathbf{p} = \boldsymbol{\tau}^T \mathbf{n}$, which, due to the symmetry of the stress tensor, can be written

as $\mathbf{p} = \boldsymbol{\tau}\mathbf{n}$. Hence, any plane on which the traction is purely normal must satisfy the equation

$$\boldsymbol{\tau}\mathbf{n} = \sigma\mathbf{n}. \quad (2.45)$$

The left-hand side of (2.45) represents a matrix, $\boldsymbol{\tau}$, multiplying a vector, $\mathbf{n}$, whereas on the right-hand side the vector $\mathbf{n}$ is multiplied by a scalar, σ . If the 2×2 identity matrix is denoted by $\mathbf{I}$, then $\mathbf{n} = \mathbf{I}\mathbf{n}$, and (2.45) can be rewritten as

$$(\boldsymbol{\tau} - \sigma\mathbf{I})\mathbf{n} = 0. \quad (2.46)$$

Equation (2.46) can be recognized as a standard eigenvalue problem, in which σ is the *eigenvalue*, and $\mathbf{n}$ is the *eigenvector*. Much of the theory of stress follows immediately from the theory pertaining to eigenvectors and eigenvalues of a symmetric matrix. The main conclusions of this theory in an arbitrary number of dimensions N are (Lang, 1971) that there will always be N mutually orthogonal eigenvectors, each corresponding to a real eigenvalue σ , although the eigenvalues need not necessarily be distinct from each other. In the present case, the eigenvalues are the principal stresses, and the associated eigenvectors are the principal stress directions. These results, along with explicit expressions for the principal stresses and principal stress directions, can be derived from (2.46) without appealing to the general theory, however, as follows.

Equation (2.46) can be written in component form as

$$(\tau_{xx} - \sigma)n_x + \tau_{xy}n_y = 0, \quad (2.47)$$

$$\tau_{yx}n_x + (\tau_{yy} - \sigma)n_y = 0. \quad (2.48)$$

Using the standard procedure of Gaussian elimination, we multiply (2.47) by τ_{yx} , and multiply (2.48) by $(\tau_{xx} - \sigma)$, to arrive at

$$(\tau_{xx} - \sigma)\tau_{yx}n_x + \tau_{xy}\tau_{yx}n_y = 0, \quad (2.49)$$

$$(\tau_{xx} - \sigma)\tau_{yx}n_x + (\tau_{yy} - \sigma)(\tau_{xx} - \sigma)n_y = 0. \quad (2.50)$$

Subtraction of (2.49) from (2.50) yields

$$[\sigma^2 - (\tau_{xx} + \tau_{yy})\sigma + (\tau_{xx}\tau_{yy} - \tau_{xy}^2)]n_y = 0, \quad (2.51)$$

where use has been made of the relationship $\tau_{yx} = \tau_{xy}$. This equation will be satisfied if either the bracketed term vanishes, or if $n_y = 0$. In this latter case, we must have $n_x = 1$, since $\mathbf{n}$ is a unit vector. Equation (2.47) then shows that $\sigma = \tau_{xx}$, and (2.48) shows that $\tau_{xy} = 0$. This solution therefore corresponds to the special case in which the x direction is already a principal stress direction, and τ_{xx} is a principal stress. In general, this will not be the case, and we must proceed by setting the bracketed term to zero:

$$\sigma^2 - (\tau_{xx} + \tau_{yy})\sigma + (\tau_{xx}\tau_{yy} - \tau_{xy}^2) = 0. \quad (2.52)$$

The bracketed term in (2.51) is the *determinant* of the matrix $(\boldsymbol{\tau} - \sigma\mathbf{I})$, so (2.52) can be written symbolically as $\det(\boldsymbol{\tau} - \sigma\mathbf{I}) = 0$, which is the standard criterion

for finding the eigenvalues of a matrix. This equation is a quadratic in σ , and will always have two roots, which will be functions of the two parenthesized coefficients that appear in (2.52). Before discussing these roots, we note that as the two principal stresses are scalars, their values should not depend on the coordinate system used. Therefore, the two coefficients $(\tau_{xx} + \tau_{yy})$ and $(\tau_{xx}\tau_{yy} - \tau_{xy}^2)$, must be independent of the coordinate system being used; this could also be shown more directly by adding (2.25) and (2.26). These two combinations of the stress components are known as *invariants*, and are discussed in more detail in a three-dimensional context in §2.8.

The quadratic formula shows that the two roots of (2.52) are given by the two values σ_1 and σ_2 from (2.37) and (2.38). If σ takes on one of these two values, (2.47) and (2.48) become linearly dependent. In this case, one of the two equations is redundant, and we can solve (2.47) to find

$$\tan \theta = \frac{n_y}{n_x} = \frac{2\tau_{xy}}{(\tau_{xx} - \tau_{yy}) \pm [4\tau_{xy}^2 + (\tau_{xx} - \tau_{yy})^2]^{1/2}}, \quad (2.53)$$

where the $+$ sign corresponds to σ_1 , and the $-$ sign corresponds to σ_2 . Using the trigonometric identity $\tan 2\theta = 2 \tan \theta / (1 - \tan^2 \theta)$, it can be shown that (2.53) is consistent with (2.33). These two directions, corresponding to the two orthogonal unit eigenvectors, will define a new coordinate system, rotated by an angle θ from the x direction, in which the shear stresses are zero. This coordinate system is often referred to as the *principal coordinate system*.

2.4 Graphical representations of stress in two dimensions

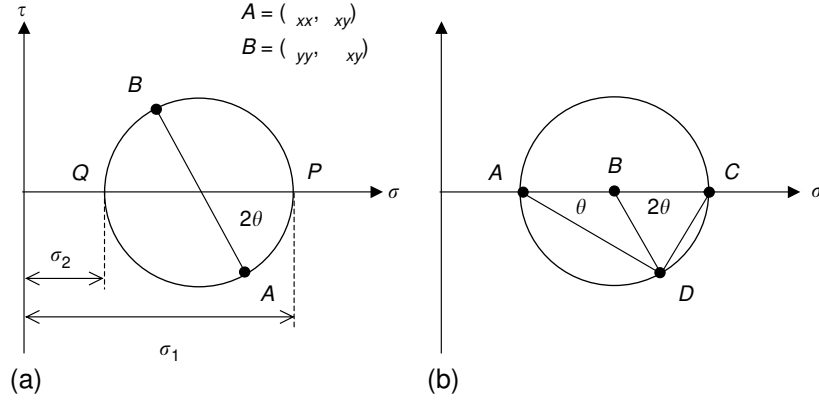
A simple graphical construction popularized by Mohr (1914) can be used to represent the state of stress at a point. Recall that (2.31) and (2.32) give expressions for the normal stress and shear stress acting on a plane whose unit normal direction is rotated from the x direction by a counterclockwise angle θ . Now imagine that we are using the principal coordinate system, in which the shear stresses are zero and the normal stresses are the two principal normal stresses. In this case we replace τ_{xx} with σ_1 , replace τ_{yy} with σ_2 , replace τ_{xy} with 0, and interpret θ as the angle of counterclockwise rotation from the direction of the maximum principal stress. We thereby arrive at the following equations that give the normal and shear stresses on a plane whose outward unit normal vector is rotated by θ from the first principal direction:

$$\sigma = \frac{(\sigma_1 + \sigma_2)}{2} + \frac{(\sigma_1 - \sigma_2)}{2} \cos 2\theta, \quad (2.54)$$

$$\tau = \frac{-(\sigma_1 - \sigma_2)}{2} \sin 2\theta. \quad (2.55)$$

These are the equations of a circle in the (σ, τ) plane, with its center at the point $\{\sigma = (\sigma_1 + \sigma_2)/2, \tau = 0\}$, and with radius $(\sigma_1 - \sigma_2)$. In contrast to the standard parameterization in which the angle θ is measured in the counterclockwise direction, this circle is parameterized in the clockwise direction, with angle 2θ . This becomes clear if we note that $\cos 2\theta$ can be replaced with $\cos(-2\theta)$ in (2.54),

Fig. 2.6 Mohr's circle construction (see text for discussion).



and $-\sin 2\theta$ can be replaced with $\sin(-2\theta)$ in (2.55). A generic Mohr's circle is shown in Fig. 2.6a. A detailed discussion of the use of Mohr's circle in rock and soil mechanics is given by Parry (1995).

Many of the important properties of the two-dimensional stress tensor can be read directly off of the Mohr's circle. For example, at point P , when $\theta = 0$, there is no rotation from the σ_1 direction, and indeed Mohr's circle indicates that $(\sigma = \sigma_1, \tau = 0)$. Similarly, consider the plane for which $\theta = 90^\circ$. This plane is rotated counterclockwise by 90° from the σ_1 direction, and therefore represents the σ_2 direction. This plane is represented on Mohr's circle by the point that is rotated clockwise by $2\theta = 180^\circ$, which is point Q on Fig. 2.6a, where we find $(\sigma = \sigma_2, \tau = 0)$. This construction also clearly shows that the maximum shear stress has a magnitude equal to the radius of the Mohr's circle, and occurs on planes for which $2\theta = \pm 90^\circ$, which is to say $\theta = \pm 45^\circ$. These two planes bisect the two planes on which the principal normal stresses act, which are $\theta = 0^\circ, 90^\circ$.

Point A on the Mohr's circle in Fig. 2.6a shows the stresses acting on a generic plane whose unit normal vector is rotated by angle θ from the σ_1 direction. This direction can be denoted as the x direction, and these stresses can therefore be denoted by $(\sigma = \tau_{xx}, \tau = \tau_{xy})$. Now consider the plane that is rotated by an additional 90° . For this plane, the additional increment in 2θ is 180° , and the stresses are represented by the point B , which is located at the opposite end of a diameter of the circle from point A . This direction can be denoted as the y direction, in which case the x and y directions define an orthogonal coordinate system. However, the stresses at point B on Mohr's circle must be identified as $(\sigma = \tau_{yy}, \tau = -\tau_{yx})$. This is because it is implicit in (2.55) that the tangential direction is rotated 180° counterclockwise from the normal direction of the plane in question, which would then correspond to the $-x$ direction instead of the $+x$ direction.

It is also seen from Mohr's circle that the mean value of the two normal stresses, $(\tau_{xx} + \tau_{yy})/2$, is equal to the horizontal distance from the origin to the center of Mohr's circle, which is $(\sigma_1 + \sigma_2)/2$. This is another proof of the fact that the value of the mean normal stress is independent of the coordinate system used.

Mohr's circle can also be used to graphically determine the two principal stresses, and the orientations of the principal stress directions, given knowledge of the components of the stress tensor in some (x, y) coordinate system. We first plot the point (τ_{xx}, τ_{xy}) on the (σ, τ) plane, and note that these two stresses will be the normal and shear stresses on the plane whose outward unit normal vector is $\mathbf{e}_x$. This direction is rotated by some (as yet unknown) angle θ from the σ_1 direction. We next plot the stresses $(\tau_{yy}, -\tau_{yx})$ on the (σ, τ) plane, and note that these represent the stresses on the plane with outward unit normal vector $\mathbf{e}_y$. This direction is therefore rotated by an angle $\theta + 90^\circ$ from the σ_1 direction. In accordance with the earlier discussion, the sign convention that is used for the shear stress on this second plane on a Mohr's diagram is opposite to that used when considering this as the second direction in an orthogonal coordinate system; hence, this second pair is plotted as $(\tau_{yy}, -\tau_{yx})$. As these two planes are rotated from one another by 90° , they will be separated by 180° on Mohr's circle; hence, the line joining these two points will be the diameter of Mohr's circle. Once this diameter is constructed, the circle can be drawn with a compass. The two points at which this circle intersects the σ -axis will be the two principal stresses, σ_1 and σ_2 . The angle of rotation between the x direction and the σ_1 direction can also be read directly from this circle.

Mohr's circle can also be used to graphically find the orientation of the plane on which certain tractions act (Kuske and Robertson, 1974). Consider point D in Fig. 2.6b, at which the traction is given by (σ, τ) . First note that $\angle DBA = \pi - \angle DBC$. Next, note that $\angle DAB$ and $\angle ADB$ are two equal angles of an isosceles triangle, the third angle of which is $\angle DBA$. It follows that $\angle DAB = \theta$. The chord AD therefore points in the direction of the outward unit normal vector to the plane in question. Since $\angle ADC$ is inscribed within a semicircle, we know that $\angle ADC = \pi/2$. Chords AD and DC are therefore perpendicular to each other, from which it follows that chord CD indicates the direction of the plane on which the tractions are (σ, τ) . This construction is sometimes useful in aiding in the visualization of the tractions acting on various planes.

There are other geometrical constructions that have been devised to represent the state of stress at a point in a body. Most of these are less convenient than Mohr's circle, and to a great extent these graphical approaches, once very popular, have been superseded by algebraic methods. Nevertheless, we briefly mention Lamé's stress ellipsoid, which in two dimensions is a stress ellipse. To simplify the discussion, assume that we are using the principal coordinate system, in which case it follows from (2.9) and (2.10) that

$$p_1(\mathbf{n}) = \sigma_1 n_1 \quad \text{and} \quad p_2(\mathbf{n}) = \sigma_2 n_2, \quad (2.56)$$

where we have let $x \rightarrow 1, y \rightarrow 2$, and have noted that, by construction, $\tau_{12} = 0$. Since $\mathbf{n}$ is a unit vector, we see from (2.56) that

$$(p_1/\sigma_1)^2 + (p_2/\sigma_2)^2 = (n_1)^2 + (n_2)^2 = 1. \quad (2.57)$$

The point (p_1, p_2) therefore traces out an ellipse whose semimajor and semiminor axes are σ_1 and σ_2 , respectively (Fig. 2.7). Each vector from the origin to a point

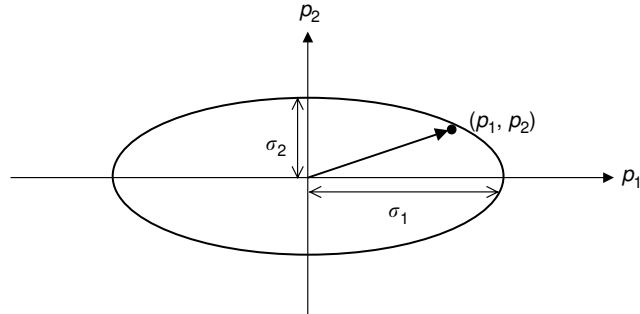


Fig. 2.7 Lamé's stress ellipse (see text for description).

on the ellipse represents a traction vector that acts on some plane passing through the point at which the principal stresses are σ_1 and σ_2 . However, although the Lamé stress ellipse shows the various traction vectors that act on different planes, it does *not* indicate the plane on which the given traction acts. In general, only when the vector OP lies along one of the principal directions in Fig. 2.7 will the direction of the plane be apparent, since in these special cases the traction is known to be normal to the plane. In the more general case, the direction of the unit normal vector of the plane on which the traction is (p_1, p_2) can be found with the aid of the stress-director surface, which is defined by

$$(p_1^2/\sigma_1) + (p_2^2/\sigma_2) = \pm 1. \quad (2.58)$$

For the case which is most common in rock mechanics, in which both principal stresses are positive, the $+$ sign must be used in (2.58), and the surface is an ellipse with axes $\sqrt{\sigma_1}$ and $\sqrt{\sigma_2}$. The outward unit normal vector of the plane on which the traction is (p_1, p_2) is then given by the tangent to the stress-director ellipse at the point where it intersects the stress ellipsoid (Chou and Pagano, 1992, p. 200). Proof of this assertion, and more details of this construction, can be found in Timoshenko and Goodier (1970) and Durelli et al. (1958).

One interesting fact that is more apparent from the Lamé construction than from Mohr's circle is that not only does the magnitude of the normal component of the stress take on stationary values in the principal directions, but the magnitude of the *total* traction vector also takes on stationary values in these directions. In particular, the maximum value of $|\mathbf{p}|$ is seen to be equal to σ_1 , and occurs in the direction of the major principal stress.

Most of the manipulations and transformations described above are concerned with the values of the stress and traction at a given "point" in the rock. In general, the state of stress will vary from point to point. The equations that govern these variations are described in §5.5. The state of stress in a rock mass can either be estimated based on a solution (either numerical or analytical) of these equations (Chapter 8), or from stress measurements (Chapter 13). In order to completely specify the state of stress in a two-dimensional rock mass, it is necessary either to know the values of τ_{xx} , τ_{yy} , and τ_{xy} at each point in the body, or, alternatively, to know at each point the values of the two principal stresses σ_1 and σ_2 , along with the angle of inclination between

the x direction and, say, the σ_1 direction. Although it is difficult to display all of this information graphically, there are a number of simple graphical representations that are useful in giving a partial picture of a stress field. Among these are:

- 1 *Isobars*, which are curves along which the principal stress is constant. There are two sets of isobars, one for σ_1 and one for σ_2 . A set of isobars for one of the principal stresses, say σ_1 , must by definition form a nonintersecting set of curves. However, an isobar of σ_1 may intersect an isobar of σ_2 .
- 2 *Isochromatics*, which are curves along which the maximum shear stress $(\sigma_1 - \sigma_2)/2$, is constant. These curves can be directly found using the methods of photoelasticity, which is described by Frocht (1941) and Durelli et al. (1958).
- 3 *Isopachs*, which are curves along which the mean normal stress $(\sigma_1 + \sigma_2)/2$ is constant. It is shown in §5.5 that this quantity satisfies Laplace's equation, which is the same equation that governs, for example, steady-state temperature distributions, or steady-state distributions of the electric field, in isotropic conducting bodies. Hence, the isopachs can be found from analogue methods that utilize electrically conducting paper that is cut to the same shape as the rock mass under investigation. This procedure is discussed by Durelli et al. (1958).
- 4 *Isostatics*, or *stress trajectories*, are a system of curves which are at each point tangent to the principal axes of the stress. As the two principal axes are always orthogonal, the two sets of isostatic curves are mutually orthogonal. Since a free surface is always a principal plane (as it has no shear stress acting on it), an isostatic curve will intersect a free surface at a right angle to it.
- 5 *Isoclinics*, which are curves on which the principal axes make a constant angle with a given fixed reference direction. These curves can also be obtained by photoelastic methods.
- 6 *Slip lines*, which are curves on which the shear stress is a maximum. As the maximum shear stress at any point is always in a direction that bisects the two directions of principal normal stresses, these lines form an orthogonal grid.

2.5 Stresses in three dimensions

The theory of stresses in three dimensions is in general a straightforward extension of the two-dimensional theory. A generic plane in three dimensions will have a unit normal vector $\mathbf{n} = (n_x, n_y, n_z)$. The components of this vector satisfy the normalization condition $(n_x)^2 + (n_y)^2 + (n_z)^2 = 1$. A three-dimensional version of the argument accompanying Fig. 2.2b leads to the following generalization of (2.6):

$$\mathbf{p}(\mathbf{n}) = n_x \mathbf{p}(\mathbf{e}_x) + n_y \mathbf{p}(\mathbf{e}_y) + n_z \mathbf{p}(\mathbf{e}_z). \quad (2.59)$$

The components of the three traction vectors that act on planes whose outward unit normals are in the three coordinate directions are denoted by

$$\mathbf{p}(\mathbf{e}_x) = [\tau_{xx} \quad \tau_{xy} \quad \tau_{xz}]^T, \quad (2.60)$$

$$\mathbf{p}(\mathbf{e}_y) = [\tau_{yx} \quad \tau_{yy} \quad \tau_{yz}]^T, \quad (2.61)$$

$$\mathbf{p}(\mathbf{e}_z) = [\tau_{zx} \quad \tau_{zy} \quad \tau_{zz}]^T. \quad (2.62)$$

Substitution of (2.60)–(2.62) into (2.59) leads to

$$p_x(\mathbf{n}) = \tau_{xx}n_x + \tau_{yx}n_y + \tau_{zx}n_z, \quad (2.63)$$

$$p_y(\mathbf{n}) = \tau_{xy}n_x + \tau_{yy}n_y + \tau_{zy}n_z, \quad (2.64)$$

$$p_z(\mathbf{n}) = \tau_{xz}n_x + \tau_{yz}n_y + \tau_{zz}n_z, \quad (2.65)$$

which can be written in matrix form as $\mathbf{p}(\mathbf{n}) = \boldsymbol{\tau}^T \mathbf{n}$, that is,

$$\begin{bmatrix} p_x(\mathbf{n}) \\ p_y(\mathbf{n}) \\ p_z(\mathbf{n}) \end{bmatrix} = \begin{bmatrix} \tau_{xx} & \tau_{yx} & \tau_{zx} \\ \tau_{xy} & \tau_{yy} & \tau_{zy} \\ \tau_{xz} & \tau_{yz} & \tau_{zz} \end{bmatrix} \begin{bmatrix} n_x \\ n_y \\ n_z \end{bmatrix}. \quad (2.66)$$

The three-dimensional analogue of the argument illustrated by Fig. 2.3b would show that the conjugate terms in the three-dimensional stress tensor are equal, that is,

$$\tau_{yx} = \tau_{xy}, \quad \tau_{yz} = \tau_{zy}, \quad \tau_{zx} = \tau_{xz}. \quad (2.67)$$

Hence, (2.66) can also be written as $\mathbf{p}(\mathbf{n}) = \boldsymbol{\tau} \mathbf{n}$.

The question can again be asked as to whether or not there are planes on which the shear stresses vanish. On such planes, the traction vector will be parallel to the outward unit normal vector, and therefore can be written as $\mathbf{p} = \sigma \mathbf{n}$, where σ is some scalar. But as $\mathbf{p}(\mathbf{n}) = \boldsymbol{\tau} \mathbf{n}$, we have $\boldsymbol{\tau} \mathbf{n} = \sigma \mathbf{n} = \sigma \mathbf{I} \mathbf{n}$, and therefore again arrive at the eigenvalue problem $(\boldsymbol{\tau} - \sigma \mathbf{I}) \mathbf{n} = \mathbf{0}$, §2.3 (2.46), that is,

$$\begin{bmatrix} \tau_{xx} - \sigma & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \tau_{yy} - \sigma & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \tau_{zz} - \sigma \end{bmatrix} \begin{bmatrix} n_x \\ n_y \\ n_z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}. \quad (2.68)$$

From this point on, the development follows that for the two-dimensional theory. Although $(n_x, n_y, n_z) = (0, 0, 0)$ is obviously a solution to (2.68), it is inadmissible because it does not satisfy the condition that $\mathbf{n} \cdot \mathbf{n} = 1$. Admissible solutions can be found only if the determinant of the matrix $(\boldsymbol{\tau} - \sigma \mathbf{I})$ vanishes (Lang, 1971). When the determinant is expanded out, it takes the form

$$\sigma^3 - I_1 \sigma^2 - I_2 \sigma - I_3 = 0, \quad (2.69)$$

where

$$I_1 = \tau_{xx} + \tau_{yy} + \tau_{zz}, \quad (2.70)$$

$$I_2 = \tau_{xy}^2 + \tau_{xz}^2 + \tau_{yz}^2 - \tau_{xx}\tau_{yy} - \tau_{xx}\tau_{zz} - \tau_{yy}\tau_{zz}, \quad (2.71)$$

$$I_3 = \tau_{xx}\tau_{yy}\tau_{zz} + 2\tau_{xy}\tau_{xz}\tau_{yz} - \tau_{xx}\tau_{yz}^2 - \tau_{yy}\tau_{xz}^2 - \tau_{zz}\tau_{xy}^2. \quad (2.72)$$

The fact that the stress tensor is symmetric ensures that (2.69) has three *real* roots. These roots are conventionally labeled such that $\sigma_1 \geq \sigma_2 \geq \sigma_3$. Each of these roots will correspond to an eigenvector that can be labeled as $\mathbf{n}^1 = (n_x^1, n_y^1, n_z^1)$, etc. Although, in general, eigenvectors are arbitrary to within a multiplicative

constant, the condition that the vector must be of unit length leads to a unique determination of the vector components. The normalized eigenvector corresponding to σ_1 is the unit normal vector of the plane on which the normal stress is σ_1 and the shear stress is 0, and likewise for the other two principal stresses.

In two dimensions it was shown explicitly in (2.33) that the two principal stress directions are rotated by 90° from each other. The orthogonality of the principal directions continues to be the case in three dimensions; this can be proven as follows. First recall that the inner product of two vectors, $\mathbf{u} \cdot \mathbf{v}$, can also be expressed as $\mathbf{u}^T \mathbf{v}$, since

$$\mathbf{u}^T \mathbf{v} = \begin{bmatrix} u_x & u_y & u_z \end{bmatrix} \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix} = u_x v_x + u_y v_y + u_z v_z = \mathbf{u} \cdot \mathbf{v}. \quad (2.73)$$

Also, recall that the transpose operation has the properties that $(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$, and $(\mathbf{A}^T)^T = \mathbf{A}$. Consider two principal stresses, $\sigma_1 \neq \sigma_2$, and the principal directions corresponding to these two stresses, that is, $\boldsymbol{\tau} \mathbf{n}^1 = \sigma_1 \mathbf{n}^1$ and $\boldsymbol{\tau} \mathbf{n}^2 = \sigma_2 \mathbf{n}^2$. Take the inner product of both sides of this first equation with $\mathbf{n}^2$:

$$(\mathbf{n}^2)^T \boldsymbol{\tau} \mathbf{n}^1 = (\mathbf{n}^2)^T \sigma_1 \mathbf{n}^1 = \sigma_1 (\mathbf{n}^2)^T \mathbf{n}^1, \quad (2.74)$$

and the inner product of both sides of the second equation with $\mathbf{n}^1$:

$$(\mathbf{n}^1)^T \boldsymbol{\tau} \mathbf{n}^2 = (\mathbf{n}^1)^T \sigma_2 \mathbf{n}^2 = \sigma_2 (\mathbf{n}^1)^T \mathbf{n}^2. \quad (2.75)$$

Now take the transpose of both sides of (2.75):

$$[(\mathbf{n}^1)^T \boldsymbol{\tau} \mathbf{n}^2]^T = (\mathbf{n}^2)^T \boldsymbol{\tau}^T \mathbf{n}^1 = [\sigma_2 (\mathbf{n}^1)^T \mathbf{n}^2]^T = \sigma_2 (\mathbf{n}^2)^T \mathbf{n}^1. \quad (2.76)$$

But $\boldsymbol{\tau}^T = \boldsymbol{\tau}$, so (2.76) shows that

$$(\mathbf{n}^2)^T \boldsymbol{\tau} \mathbf{n}^1 = \sigma_2 (\mathbf{n}^2)^T \mathbf{n}^1. \quad (2.77)$$

Subtracting (2.77) from (2.74) yields

$$0 = (\sigma_1 - \sigma_2) (\mathbf{n}^2)^T \mathbf{n}^1 = (\sigma_1 - \sigma_2) \mathbf{n}^2 \cdot \mathbf{n}^1. \quad (2.78)$$

As $\sigma_1 \neq \sigma_2$ by assumption, it follows that $\mathbf{n}^2 \cdot \mathbf{n}^1 = 0$, which is to say that $\mathbf{n}^1$ and $\mathbf{n}^2$ are orthogonal to each other. This argument applies to any pair of distinct principal stresses, proving that the three principal stresses act on three mutually orthogonal planes.

In certain cases, two of the roots of (2.69) may be equal, say $\sigma_1 > \sigma_2 = \sigma_3$. Assume that this is the case, and that we have found two vectors $\mathbf{n}^2$ and $\mathbf{n}^3$ that both satisfy $\boldsymbol{\tau} \mathbf{n} = \sigma_2 \mathbf{n}$. Any vector $\mathbf{n}$ in the plane that is spanned by these two vectors can be written in the form $\mathbf{n} = c_2 \mathbf{n}^2 + c_3 \mathbf{n}^3$, so

$$\begin{aligned} \boldsymbol{\tau} \mathbf{n} &= \boldsymbol{\tau} (c_2 \mathbf{n}^2 + c_3 \mathbf{n}^3) = c_2 \boldsymbol{\tau} \mathbf{n}^2 + c_3 \boldsymbol{\tau} \mathbf{n}^3 = c_2 \sigma_2 \mathbf{n}^2 + c_3 \sigma_2 \mathbf{n}^3 \\ &= \sigma_2 (c_2 \mathbf{n}^2 + c_3 \mathbf{n}^3) = \sigma_2 \mathbf{n}, \end{aligned} \quad (2.79)$$

which proves that $\mathbf{n}$ is also an eigenvector of $\boldsymbol{\tau}$, with eigenvalue σ_2 . Hence, all vectors in the plane spanned by $\mathbf{n}^2$ and $\mathbf{n}^3$ are principal stress directions associated with σ_2 . No more than two of these vectors can be linearly independent, however, and it is conventional to pick two *orthogonal* directions in this plane as the principal stress directions associated with σ_2 . In this way we maintain the orthogonality of the principal stress directions. In the special case in which all three principal stresses are equal, that is, $\sigma_1 = \sigma_2 = \sigma_3$, the same argument shows that *any* vector of unit length will be an eigenvector corresponding to σ_1 , and will therefore define a principal stress direction. In other words, the traction on *all* planes will be a normal traction of magnitude σ_1 .

In two dimensions, (2.39) showed that the principal stresses were also the maximum and minimum values of the normal stress that acted on any plane. This also follows directly from Mohr's circle, since the two locations at which the circle intersects the $\tau = 0$ axis are the extreme values of σ . In three dimensions, the three principal stresses also represent locally stationary values of σ , one of which is an absolute maximum, one an absolute minimum, and one a saddle point in the three-dimensional space of unit vectors $\mathbf{n}$. To prove this, we start with the fact that the normal component of the traction, p_n , can be found from

$$p_n = \mathbf{p} \cdot \mathbf{n} = \mathbf{n} \cdot \mathbf{p} = \mathbf{n}^T \mathbf{p} = \mathbf{n}^T \boldsymbol{\tau} \mathbf{n}. \quad (2.80)$$

By performing the indicated matrix multiplication, or by analogy with (2.19), we find

$$p_n = \tau_{xx}n_x^2 + \tau_{yy}n_y^2 + \tau_{zz}n_z^2 + 2\tau_{xy}n_xn_y + 2\tau_{xz}n_xn_z + 2\tau_{yz}n_yn_z. \quad (2.81)$$

It is now desired to find the extreme values of p_n , as a function of the three components of the unit normal vector $\mathbf{n}$, bearing in mind the constraint $n_x^2 + n_y^2 + n_z^2 = 1$. This constraint equation is of the form $f(n_x, n_y, n_z) = \text{constant}$. Geometrically, this is equivalent to considering the variation of p_n on the surface of the unit sphere in (n_x, n_y, n_z) space. According to the theory of Lagrange multipliers (Lang, 1973, pp. 140–4), the constrained maximum (or minimum) will occur at a point at which the gradient of p_n is parallel to the gradient of the constraint function f . The components of the gradient of p_n , considered as a function of (n_x, n_y, n_z) , are calculated as follows:

$$\frac{\partial p_n}{\partial n_x} = 2\tau_{xx}n_x + 2\tau_{xy}n_y + 2\tau_{xz}n_z = 2(\tau_{xx}n_x + \tau_{xy}n_y + \tau_{xz}n_z) = 2p_x, \quad (2.82)$$

and similarly for the other two components. Hence, the gradient of p_n is

$$\text{grad}(p_n) = \left(\frac{\partial p_n}{\partial n_x}, \frac{\partial p_n}{\partial n_y}, \frac{\partial p_n}{\partial n_z} \right) = 2(p_x, p_y, p_z) = 2\mathbf{p}. \quad (2.83)$$

The gradient of the constraint function is

$$\text{grad}(f) = \text{grad}(n_x^2 + n_y^2 + n_z^2) = (2n_x, 2n_y, 2n_z) = 2\mathbf{n}. \quad (2.84)$$

If these two gradients are parallel, then $2\mathbf{p} = \sigma(2\mathbf{n})$ for some constant σ , which is to say $\mathbf{p} = \sigma\mathbf{n}$. But this is precisely the condition that defines the principal

stresses and their associated principal directions. Hence, the principal directions also define those planes on which the normal traction takes on stationary values.

It is also of some interest to find the stationary values of the shear traction, and the planes on which these tractions act. For this purpose, it is convenient to work in the principal coordinate system. The traction vector on an arbitrary plane whose unit normal vector (in the principal coordinate system) is $\mathbf{n} = (n_1, n_2, n_3)$ is found from (2.66) to be

$$\mathbf{p} = \boldsymbol{\tau}\mathbf{n} = \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{bmatrix} \begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix} = \begin{bmatrix} \sigma_1 n_1 \\ \sigma_2 n_2 \\ \sigma_3 n_3 \end{bmatrix}. \quad (2.85)$$

The normal traction on this plane, σ , is found by projecting $\mathbf{p}$ onto $\mathbf{n}$:

$$\sigma = \mathbf{p} \cdot \mathbf{n} = \mathbf{p}^T \mathbf{n} = \begin{bmatrix} \sigma_1 n_1 & \sigma_2 n_2 & \sigma_3 n_3 \end{bmatrix} \begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix} = \sigma_1 n_1^2 + \sigma_2 n_2^2 + \sigma_3 n_3^2. \quad (2.86)$$

The magnitude (squared) of the total traction vector is given by

$$|\mathbf{p}|^2 = \mathbf{p} \cdot \mathbf{p} = \mathbf{p}^T \mathbf{p} = \begin{bmatrix} \sigma_1 n_1 & \sigma_2 n_2 & \sigma_3 n_3 \end{bmatrix} \begin{bmatrix} \sigma_1 n_1 \\ \sigma_2 n_2 \\ \sigma_3 n_3 \end{bmatrix} = \sigma_1^2 n_1^2 + \sigma_2^2 n_2^2 + \sigma_3^2 n_3^2. \quad (2.87)$$

By the Pythagorean theorem,

$$\begin{aligned} \tau^2 &= |\mathbf{p}|^2 - \sigma^2 = \sigma_1^2 n_1^2 + \sigma_2^2 n_2^2 + \sigma_3^2 n_3^2 - (\sigma_1 n_1^2 + \sigma_2 n_2^2 + \sigma_3 n_3^2)^2 \\ &= (\sigma_1 - \sigma_2)^2 n_1^2 n_2^2 + (\sigma_2 - \sigma_3)^2 n_2^2 n_3^2 + (\sigma_3 - \sigma_1)^2 n_3^2 n_1^2. \end{aligned} \quad (2.88)$$

To find the local maximum and minimum values of the shear stress magnitude, we must optimize τ^2 subject to the constraint

$$f(n_1, n_2, n_3) = n_1^2 + n_2^2 + n_3^2 = 1. \quad (2.89)$$

By Lagrange's theorem, the stationary values of τ^2 occur where the gradient of τ^2 is parallel to the gradient of f ; this leads to the following three scalar equations:

$$[(\sigma_1 - \sigma_2)^2 n_2^2 + (\sigma_1 - \sigma_3)^2 n_3^2] n_1 = c n_1, \quad (2.90)$$

$$[(\sigma_1 - \sigma_2)^2 n_1^2 + (\sigma_2 - \sigma_3)^2 n_3^2] n_2 = c n_2, \quad (2.91)$$

$$[(\sigma_2 - \sigma_3)^2 n_2^2 + (\sigma_1 - \sigma_3)^2 n_1^2] n_3 = c n_3, \quad (2.92)$$

where c is a Lagrange multiplier, the value of which is not needed in the present discussion.

Equation (2.90) can be satisfied by picking $n_1 = 0$, after which (2.91) and (2.92) show that $n_3^2 = n_2^2$. Invoking the normalization constraint (2.89) then shows that

$n_3^2 = n_2^2 = 1/2$. Hence, one family of planes on which τ will take on extreme values will correspond to

$$\mathbf{n} = (0, \pm 1/\sqrt{2}, \pm 1/\sqrt{2}). \quad (2.93)$$

On these planes, (2.88) shows that $|\tau| = (\sigma_2 - \sigma_3)/2$, and (2.86) shows that $\sigma = (\sigma_2 + \sigma_3)/2$. Another set of solutions to (2.90)–(2.92) is given by

$$\mathbf{n} = (\pm 1/\sqrt{2}, 0, \pm 1/\sqrt{2}), \quad (2.94)$$

on which planes $|\tau| = (\sigma_1 - \sigma_3)/2$ and $\sigma = (\sigma_1 + \sigma_3)/2$. Finally, a third set of solutions is

$$\mathbf{n} = (\pm 1/\sqrt{2}, \pm 1/\sqrt{2}, 0), \quad (2.95)$$

on which planes $|\tau| = (\sigma_1 - \sigma_2)/2$ and $\sigma = (\sigma_1 + \sigma_2)/2$. These extreme values of the shear stress are denoted by τ_1 , τ_2 , and τ_3 , respectively. As the shear stresses are zero on the planes of the principal normal stresses, the extreme values found above are local maxima.

Thus far, we have found the magnitudes of the maximum shear stresses and the unit normals of the planes on which they act. To find the *directions within those planes* in which the shears act, consider one of these planes, and let $\mathbf{t}$ be the unit vector pointing in the direction of this shear traction. We decompose the total traction vector into a shear and a normal component, as follows:

$$\mathbf{p} = \sigma \mathbf{n} + \tau \mathbf{t}. \quad (2.96)$$

Solving (2.96) for $\mathbf{t}$, and making use of (2.85), yields

$$\mathbf{t} = \frac{1}{\tau}(\mathbf{p} - \sigma \mathbf{n}) = [(\sigma_1 - \sigma)n_1/\tau \quad (\sigma_2 - \sigma)n_2/\tau \quad (\sigma_3 - \sigma)n_3/\tau]^T, \quad (2.97)$$

in which τ , σ , and $\mathbf{n}$ are one set of the solutions to (2.90)–(2.92) described above. For example, on the plane whose unit normal vector is given by $\mathbf{n} = (0, 1/\sqrt{2}, 1/\sqrt{2})$, we have $\tau = (\sigma_2 - \sigma_3)/2$ and $\sigma = (\sigma_2 + \sigma_3)/2$, and (2.97) yields $\mathbf{t} = (0, 1/\sqrt{2}, -1/\sqrt{2})$.

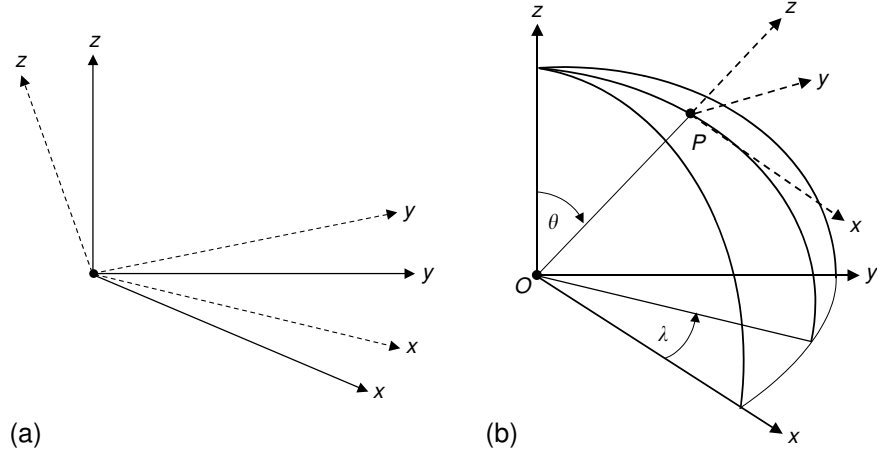
2.6 Stress transformations in three dimensions

If the stress components are known in a given coordinate system, the stress components in a second coordinate system that is rotated from the first one can be found by matrix multiplication, as in (2.30). This was proven in a two-dimensional context in §2.3, and will now be generalized to three dimensions.

Consider a coordinate system (x, y, z) , as in Fig. 2.8a, and another coordinate system (x', y', z') that is rotated by some angle in three-dimensional space. Let the direction cosines of the unit vector $\mathbf{e}_{x'}$, relative to the *unprimed* coordinate system, be (l_{11}, l_{12}, l_{13}) , and similarly for the other two unit vectors $\mathbf{e}_{y'}$ and $\mathbf{e}_{z'}$, that is,

$$\mathbf{e}_{x'} = (l_{11}, l_{12}, l_{13}), \quad \mathbf{e}_{y'} = (l_{21}, l_{22}, l_{23}), \quad \mathbf{e}_{z'} = (l_{31}, l_{32}, l_{33}). \quad (2.98)$$

Fig. 2.8 (a) Original (unprimed) coordinate system, along with another rotated (primed) coordinate system, and (b) a system that utilizes the zenith and longitudinal angles.



Now consider the traction on the plane whose outward unit normal vector is $\mathbf{e}_{x'}$. By (2.66), this traction is given by $\mathbf{p}(\mathbf{e}_{x'}) = \boldsymbol{\tau} \mathbf{e}_{x'}$, where $\boldsymbol{\tau}$ is the stress tensor in the unprimed coordinate system. The component of this traction in the $\mathbf{e}_{x'}$ direction, which by definition is $\tau_{x'x'}$, is found, as in (2.6), by taking the inner product of $\mathbf{p}(\mathbf{e}_{x'})$ with $\mathbf{e}_{x'}$, that is,

$$\begin{aligned} \tau_{x'x'} &= \mathbf{e}_{x'} \cdot \mathbf{p}(\mathbf{e}_{x'}) = (\mathbf{e}_{x'})^T \mathbf{p}(\mathbf{e}_{x'}) = (\mathbf{e}_{x'})^T \boldsymbol{\tau} \mathbf{e}_{x'} \\ &= \begin{bmatrix} l_{11} & l_{12} & l_{13} \end{bmatrix} \begin{bmatrix} \tau_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \tau_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \tau_{zz} \end{bmatrix} \begin{bmatrix} l_{11} \\ l_{12} \\ l_{13} \end{bmatrix} \\ &= l_{11}^2 \tau_{xx} + l_{11} l_{12} \tau_{xy} + l_{11} l_{13} \tau_{xz} + l_{12} l_{11} \tau_{yx} + l_{12}^2 \tau_{yy} \\ &\quad + l_{12} l_{13} \tau_{yz} + l_{13} l_{11} \tau_{zx} + l_{13} l_{12} \tau_{zy} + l_{13}^2 \tau_{zz}. \end{aligned} \quad (2.99)$$

Although use of the symmetry properties of the stress tensor would simplify (2.99) slightly, it would obscure the algebraic structure of the transformation equations, and is not introduced at this point. The other eight components of the stress tensor in the primed coordinate system can be found in the same manner; for example, $\tau_{x'y'} = \mathbf{e}_{y'} \cdot \mathbf{p}(\mathbf{e}_{x'})$, etc. The resulting equations, each of the form (2.99), can then be written in matrix form as

$$\begin{bmatrix} \tau_{x'x'} & \tau_{x'y'} & \tau_{x'z'} \\ \tau_{y'x'} & \tau_{y'y'} & \tau_{y'z'} \\ \tau_{z'x'} & \tau_{z'y'} & \tau_{z'z'} \end{bmatrix} = \begin{bmatrix} l_{11} & l_{12} & l_{13} \\ l_{21} & l_{22} & l_{23} \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} \tau_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \tau_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \tau_{zz} \end{bmatrix} \begin{bmatrix} l_{11} & l_{21} & l_{31} \\ l_{12} & l_{22} & l_{32} \\ l_{13} & l_{23} & l_{33} \end{bmatrix}. \quad (2.100)$$

This can be verified by carrying out the matrix multiplications indicated in (2.100), and comparing the results to (2.99). Equation (2.100) can be written symbolically as

$$\boldsymbol{\tau}' = \mathbf{L} \boldsymbol{\tau} \mathbf{L}^T, \quad (2.101)$$

where the rows of the rotation matrix $\mathbf{L}$ are formed from the direction cosines of the three primed unit vectors.

The components of the stress tensor in the primed coordinate system can be written out explicitly as follows, wherein the symmetry properties of the stresses are now used in order to simplify the expressions:

$$\tau_{x'x'} = l_{11}^2 \tau_{xx} + l_{12}^2 \tau_{yy} + l_{13}^2 \tau_{zz} + 2l_{11}l_{12} \tau_{xy} + 2l_{11}l_{13} \tau_{xz} + 2l_{12}l_{13} \tau_{yz}, \quad (2.102)$$

$$\tau_{y'y'} = l_{21}^2 \tau_{xx} + l_{22}^2 \tau_{yy} + l_{23}^2 \tau_{zz} + 2l_{21}l_{22} \tau_{xy} + 2l_{21}l_{23} \tau_{xz} + 2l_{22}l_{23} \tau_{yz}, \quad (2.103)$$

$$\tau_{z'z'} = l_{31}^2 \tau_{xx} + l_{32}^2 \tau_{yy} + l_{33}^2 \tau_{zz} + 2l_{31}l_{32} \tau_{xy} + 2l_{31}l_{33} \tau_{xz} + 2l_{32}l_{33} \tau_{yz}, \quad (2.104)$$

$$\begin{aligned} \tau_{x'y'} &= l_{11}l_{21} \tau_{xx} + l_{12}l_{22} \tau_{yy} + l_{13}l_{23} \tau_{zz} + (l_{11}l_{22} + l_{12}l_{21}) \tau_{xy} \\ &\quad + (l_{12}l_{23} + l_{13}l_{22}) \tau_{yz} + (l_{11}l_{23} + l_{13}l_{21}) \tau_{xz} \end{aligned} \quad (2.105)$$

$$\begin{aligned} \tau_{y'z'} &= l_{21}l_{31} \tau_{xx} + l_{22}l_{32} \tau_{yy} + l_{23}l_{33} \tau_{zz} + (l_{21}l_{32} + l_{22}l_{31}) \tau_{xy} \\ &\quad + (l_{22}l_{33} + l_{23}l_{32}) \tau_{yz} + (l_{21}l_{33} + l_{23}l_{31}) \tau_{xz} \end{aligned} \quad (2.106)$$

$$\begin{aligned} \tau_{x'z'} &= l_{11}l_{31} \tau_{xx} + l_{12}l_{32} \tau_{yy} + l_{13}l_{33} \tau_{zz} + (l_{11}l_{32} + l_{12}l_{31}) \tau_{xy} \\ &\quad + (l_{12}l_{33} + l_{13}l_{32}) \tau_{yz} + (l_{11}l_{33} + l_{13}l_{31}) \tau_{xz} \end{aligned} \quad (2.107)$$

It follows from these equations that the stress components in any particular plane transform according to the two-dimensional transformation laws of §2.3. For example, consider a rotation about the z -axis, which corresponds to a rotation matrix in which $l_{13} = l_{23} = l_{31} = l_{32} = 0$, and $l_{33} = 1$. Equations (2.102), (2.103), and (2.105) then essentially reduce to (2.107)–(2.109), aside from differences in notation. The maximum and minimum normal stresses in such a plane are often called the *subsidiary principal stresses*. Except in the special case in which z is a principal direction, these two subsidiary principal stresses will not correspond to two of the actual three-dimensional principal stresses, however.

If the unprimed coordinate system is the coordinate system of principal stresses, (2.102)–(2.107) take the form

$$\tau_{x'x'} = l_{11}^2 \tau_{xx} + l_{12}^2 \tau_{yy} + l_{13}^2 \tau_{zz}, \quad (2.108)$$

$$\tau_{y'y'} = l_{21}^2 \tau_{xx} + l_{22}^2 \tau_{yy} + l_{23}^2 \tau_{zz}, \quad (2.109)$$

$$\tau_{z'z'} = l_{31}^2 \tau_{xx} + l_{32}^2 \tau_{yy} + l_{33}^2 \tau_{zz}, \quad (2.110)$$

$$\tau_{x'y'} = l_{11}l_{21} \tau_{xx} + l_{12}l_{22} \tau_{yy} + l_{13}l_{23} \tau_{zz}, \quad (2.111)$$

$$\tau_{y'z'} = l_{21}l_{31} \tau_{xx} + l_{22}l_{32} \tau_{yy} + l_{23}l_{33} \tau_{zz}, \quad (2.112)$$

$$\tau_{x'z'} = l_{11}l_{31} \tau_{xx} + l_{12}l_{32} \tau_{yy} + l_{13}l_{33} \tau_{zz}. \quad (2.113)$$

In some situations, it is more convenient to use the longitude angle, λ , and zenith angle, θ , to specify a plane, rather than the direction cosines of that plane relative to a particular Cartesian coordinate system (Fig. 2.8b). The axes associated with the primed coordinate system are Pz' , which is in the radial direction; Px' , which is in the plane OPz and is associated with the angle θ ; and Py' , which is chosen so as to complete the right-handed coordinate system, and which points in the direction of increasing λ . The components of these unit vectors, relative to the unprimed coordinate system, will be given by (2.98),

which we now express in terms of λ and θ . It can be seen directly from Fig. 2.8b that the direction cosines of Pz' are given by

$$\mathbf{e}_{z'} = (\sin \theta \cos \lambda, \sin \theta \sin \lambda, \cos \theta). \quad (2.114)$$

The components of the unit vector through O parallel to Px' can be found by replacing θ with $\theta + (\pi/2)$, which leads to

$$\mathbf{e}_{x'} = (\cos \theta \cos \lambda, \cos \theta \sin \lambda, -\sin \theta). \quad (2.115)$$

Finally, a unit vector through O parallel to Py' is perpendicular to Oz , and makes an angle $\lambda + (\pi/2)$ with Ox , and an angle λ with Oy . Therefore,

$$\mathbf{e}_{y'} = (-\sin \lambda, \cos \lambda, 0). \quad (2.116)$$

If the axes (x, y, z) are the principal axes, then the stress components on the plane $Px'y'$ are found from (2.112), (2.114), and (2.115) to be given by

$$\tau_{z'z'} = [\sigma_1 \cos^2 \lambda + \sigma_2 \sin^2 \lambda] \sin^2 \theta + \sigma_3 \cos^2 \theta, \quad (2.117)$$

$$\tau_{y'z'} = -\frac{1}{2}(\sigma_1 - \sigma_2) \sin \theta \sin 2\lambda, \quad (2.118)$$

$$\tau_{x'z'} = \frac{1}{2}[\sigma_1 \cos^2 \lambda + \sigma_2 \sin^2 \lambda - \sigma_3] \sin 2\theta. \quad (2.119)$$

These relations have been used, for example, by Bott (1959) for defining tectonic regimes.

2.7 Mohr's representation of stress in three dimensions

Mohr's circle representation of two-dimensional stress states, described in §2.4, can be also be used in three dimensions. To simplify the notation, we denote the components of the unit normal vector of an arbitrary plane, relative to the principal coordinate system, by (l, m, n) , rather than (n_1, n_2, n_3) , and we denote the three principal directions by (x, y, z) . By (2.86) and (2.88), the normal and shear tractions acting on this plane are

$$\sigma = l^2 \sigma_1 + m^2 \sigma_2 + n^2 \sigma_3, \quad (2.120)$$

$$\tau^2 = l^2 \sigma_1 + m^2 \sigma_2 + n^2 \sigma_3 - \sigma^2, \quad (2.121)$$

where

$$l^2 + m^2 + n^2 = 1. \quad (2.122)$$

Solving these equations for the direction cosines yields

$$l^2 = \frac{(\sigma_2 - \sigma)(\sigma_3 - \sigma) + \tau^2}{(\sigma_2 - \sigma_1)(\sigma_3 - \sigma_1)}, \quad (2.123)$$

$$m^2 = \frac{(\sigma_3 - \sigma)(\sigma_1 - \sigma) + \tau^2}{(\sigma_3 - \sigma_2)(\sigma_1 - \sigma_2)}, \quad (2.124)$$

$$n^2 = \frac{(\sigma_1 - \sigma)(\sigma_2 - \sigma) + \tau^2}{(\sigma_1 - \sigma_3)(\sigma_2 - \sigma_3)}. \quad (2.125)$$

Suppose that the direction cosine n is fixed, so that the normal vector to the plane makes a fixed angle $\theta = \arccos(n)$ with the z -axis, as in Fig. 2.9a. The intersection of the normal vector with the unit sphere will lie on the small circle $F'E'D'$. Equation (2.125) can be rearranged to yield

$$\tau^2 + \left[\sigma - \frac{1}{2}(\sigma_1 + \sigma_2) \right]^2 = \frac{1}{4}(\sigma_1 - \sigma_2) + n^2(\sigma_1 - \sigma_3)(\sigma_2 - \sigma_3), \quad (2.126)$$

which is the equation of a circle in the (σ, τ) plane. The center of this circle, A , is located at

$$\sigma = \frac{1}{2}(\sigma_1 + \sigma_2), \quad \tau = 0, \quad (2.127)$$

and the radius of this circle is

$$r = \left[\frac{1}{4}(\sigma_1 - \sigma_2) + n^2(\sigma_1 - \sigma_3)(\sigma_2 - \sigma_3) \right]^{1/2}. \quad (2.128)$$

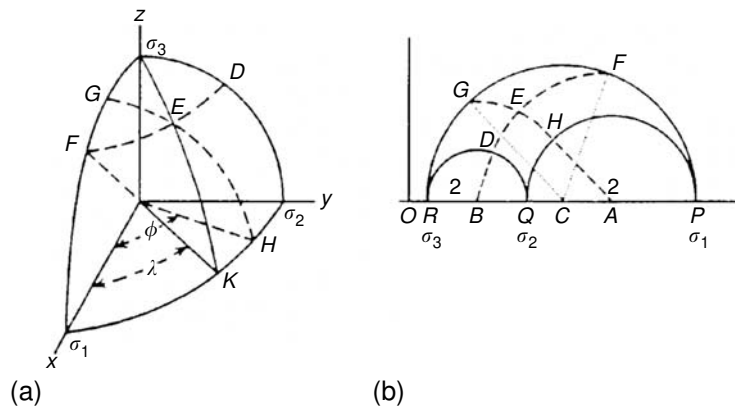
As n varies from 0 to 1, the radius varies from $AQ = (\sigma_1 - \sigma_2)/2$ to $AR = [(\sigma_1 + \sigma_2)/2] - \sigma_3$. A typical circle for an intermediate value of n is shown in Fig. 2.9b as DEF .

In the same manner, holding l constant in (2.123) gives the family of circles

$$\tau^2 + \left[\sigma - \frac{1}{2}(\sigma_2 + \sigma_3) \right]^2 = \frac{1}{4}(\sigma_2 - \sigma_3) + l^2(\sigma_2 - \sigma_1)(\sigma_3 - \sigma_1). \quad (2.129)$$

The centers of these circles are at B , located at $\sigma = (\sigma_2 + \sigma_3)/2, \tau = 0$. The radii vary from $BQ = (\sigma_2 - \sigma_3)/2$ when $l = 0$, to $BP = \sigma_1 - [(\sigma_2 + \sigma_3)/2]$ when $l = 1$. A typical circle for intermediate values of l is GEH . In Fig. 2.9b, planes for which $l = \text{constant}$ lie on a cone that makes an angle $\phi = \arccos(l)$ with the x -axis, and which intersects the unit sphere at $G'E'H'$.

Fig. 2.9 Mohr's circle in three dimensions (see text for description).



Finally, holding m constant in (2.124) gives the family of circles

$$\tau^2 + \left[\sigma - \frac{1}{2}(\sigma_1 + \sigma_3) \right]^2 = \frac{1}{4}(\sigma_1 - \sigma_3) + m^2(\sigma_3 - \sigma_2)(\sigma_1 - \sigma_2). \quad (2.130)$$

The centers of these circles are at C , located at $\sigma = (\sigma_1 + \sigma_3)/2$, $\tau = 0$. The radii vary from $CR = (\sigma_1 - \sigma_3)/2$, when $m = 0$, to $CQ = [(\sigma_1 + \sigma_3)/2] - \sigma_2$, when $m = 1$. These circles are not needed for the following analysis, and so are not shown in Fig. 2.9b.

Now consider a generic point on the unit sphere in Fig. 2.9a, such as E' , which has direction cosines $l = \cos \phi$ and $n = \cos \theta$. The traction on this plane is indicated by the point E in Fig. 2.9b, which is at the intersection of the arc DEF defined by (2.126) with $n = \cos \theta$, and the arc GEH defined by (2.129) with $l = \cos \phi$. The points at which these two arcs intersect the two smaller circles in Fig. 2.9b are found as follows. The intersection point D corresponds to the point D' in Fig. 2.9a, for which $l = 0$, $m = \sin \theta$ and $n = \cos \theta$. Hence, by (2.120) and (2.121),

$$\sigma(D) = \sigma_2 \sin^2 \theta + \sigma_3 \cos^2 \theta = \frac{1}{2}(\sigma_2 + \sigma_3) - \frac{1}{2}(\sigma_2 - \sigma_3) \cos 2\theta, \quad (2.131)$$

$$|\tau(D)| = [\sigma_2^2 \sin^2 \theta + \sigma_3^2 \cos^2 \theta - \sigma^2]^{1/2} = \frac{1}{2}(\sigma_2 - \sigma_3) \sin 2\theta. \quad (2.132)$$

But (2.131) and (2.132) are essentially the equations of Mohr's circle in the (σ_1, σ_2) plane, and so it follows that $\angle RBD = 2\theta$. A similar argument shows that $\angle HAP = 2\phi$. Therefore, we can construct a diagram such as Fig. 2.10a, from which the normal and shear tractions acting on any plane can be found by locating the intersections of the proper circles.

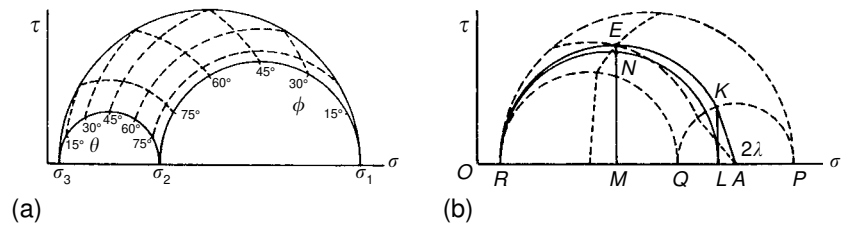
The discussion given above dealt only with the magnitude of τ , and the plane on which it acts. The line of action of the shear traction can be found by the following construction (Zizicas, 1955). Consider the latitude λ of the point E' in Fig. 2.9a. From (2.77), on the great circle for which λ is constant,

$$l^2 = (1 - n^2) \cos^2 \lambda. \quad (2.133)$$

Using (2.123) and (2.125), it can be shown that (2.133) corresponds to the following locus in the (σ, τ) plane:

$$\begin{aligned} &(\sigma_2 - \sigma_3)[(\sigma_2 - \sigma)(\sigma_3 - \sigma) + \tau^2] + (\sigma_2 - \sigma_1)[(\sigma_1 - \sigma_3)(\sigma_2 - \sigma_3) \\ &\quad - (\sigma_1 - \sigma)(\sigma_2 - \sigma) - \tau^2] \cos^2 \lambda = 0, \end{aligned} \quad (2.134)$$

Fig. 2.10 Mohr's circle in three dimensions (see text for description).



which is a circle that passes through the point $R = (\sigma_3, 0)$. When $n = 0$, which corresponds to the point K' in Fig. 2.9b, we have $l = \cos \lambda$ and $m = \sin \lambda$. It then follows from (2.120) and (2.121), and from an argument along the lines that led to (2.131) and (2.132), that

$$\sigma(K) = \frac{1}{2}(\sigma_1 + \sigma_2) + \frac{1}{2}(\sigma_1 - \sigma_2) \cos 2\lambda, \quad (2.135)$$

$$|\tau(K)| = \frac{1}{2}(\sigma_1 - \sigma_2) \sin 2\lambda. \quad (2.136)$$

Therefore, the point K lies on the circle PQ of Fig. 2.10b, and $\angle KAP = 2\lambda$. The fact that this circle passes through the points R and K , and has its center on the σ -axis, is sufficient to unambiguously define it. It must pass through the point E , which is defined by the intersection of circles GEH and DEF of Fig. 2.9b. The lines of the construction of Fig. 2.9b are shown as dotted lines in Fig. 2.10b.

Consider now the stresses in the plane $OK'Z$ of Fig. 2.9a. The normal stresses in the directions OZ and OK' are given by the lengths of the line segments OR ($= \sigma_3$) and OL in Fig. 2.10b. Therefore, the Mohr's circle for this plane has RL as its diameter. When the normal stress is OM , corresponding to the point E' , the shear stress is NM . We have thus found the component of the shear stress in the plane $OK'Z$, and the total shear stress EM must make an angle $\cos^{-1}(NM/EM)$ with that plane. This analysis gives the line of action of the shear traction, but there is still an ambiguity of sign, since our calculation has only made use of the magnitude of τ . Determination of the actual signed orientation of the shear traction, using Mohr's circle, has been discussed by Almusallam and Taher (1995).

The calculations described above, along with Figs. 2.9a and 2.10b, have been carried out using latitude and longitude coordinates, with the σ_3 -axis as the polar direction. A similar construction could be made using either of the other two principal directions as the poles.

There are numerous other applications of the Mohr's circle construction that are useful in rock and soil mechanics. Parry (1995) makes extensive use of Mohr's circle to discuss failure of geological materials. Werfel (1965) shows how to use Mohr's circle to determine the principal axes and the magnitudes of the principal stresses.

2.8 Stress invariants and stress deviation

In §2.5, the coefficients that appear in the characteristic polynomial that defines the three principal stresses were identified as the three stress invariants:

$$I_1 = \tau_{xx} + \tau_{yy} + \tau_{zz}, \quad (2.137)$$

$$I_2 = \tau_{xy}^2 + \tau_{xz}^2 + \tau_{yz}^2 - \tau_{xx}\tau_{yy} - \tau_{xx}\tau_{zz} - \tau_{yy}\tau_{zz}, \quad (2.138)$$

$$I_3 = \tau_{xx}\tau_{yy}\tau_{zz} + 2\tau_{xy}\tau_{xz}\tau_{yz} - \tau_{xx}\tau_{yz}^2 - \tau_{yy}\tau_{xz}^2 - \tau_{zz}\tau_{xy}^2. \quad (2.139)$$

The second invariant, I_2 , is sometimes defined to be the *negative* of that defined by (2.138), in which case $+I_2$, rather than $-I_2$, would appear as the coefficient of σ in (2.69). It was argued in §2.5 that these coefficients must have the same value

in all coordinate systems, since the three principal stresses, which are functions only of these three invariants, and which have a clear physical meaning that is independent of the choice of coordinate system, must themselves be invariant. A rigorous proof of this assertion is given by Spencer (1971). In fact, there are many combinations of stresses that form an invariant of the stress tensor, but all are dependent upon the set $\{I_1, I_2, I_3\}$. These three invariants can be defined in the terminology of matrix algebra by (Spencer, 1980)

$$I_1 = \text{trace}(\boldsymbol{\tau}), \quad (2.140)$$

$$I_2 = \frac{1}{2} \{ \text{trace}(\boldsymbol{\tau}^2) - [\text{trace}(\boldsymbol{\tau})]^2 \}, \quad (2.141)$$

$$I_3 = \det(\boldsymbol{\tau}), \quad (2.142)$$

where the trace operator denotes the sum of the diagonal components, and the determinant is defined in the standard manner (Lang, 1971).

Since all shear stresses, which are the off-diagonal components of the stress matrix, vanish in the principal coordinate system, the invariants take on a simpler form when expressed in that coordinate system:

$$I_1 = \sigma_1 + \sigma_2 + \sigma_3, \quad (2.143)$$

$$I_2 = -(\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_1\sigma_3), \quad (2.144)$$

$$I_3 = \sigma_1\sigma_2\sigma_3. \quad (2.145)$$

Although I_1 has exactly the same algebraic form in all coordinate systems, it would be *incorrect* to apply (2.144) or (2.145) in a nonprincipal coordinate system, since these two equations only hold when all the shear stresses are zero. For example, it is not true in general that $I_3 = \tau_{xx}\tau_{yy}\tau_{zz}$. Among the many useful relations between the stress components that can be derived with the aid of these invariants is the following:

$$\tau_{xx}^2 + \tau_{yy}^2 + \tau_{zz}^2 + 2\tau_{xy}^2 + 2\tau_{xz}^2 + 2\tau_{yz}^2 = \sigma_1^2 + \sigma_2^2 + \sigma_3^2. \quad (2.146)$$

The stress invariants are useful in the construction of stress-strain laws and failure criteria, as will be seen in Chapters 5 and 9.

Another related set of quantities that appear frequently in the development of constitutive equations for rocks is the *octahedral stresses*. These are the normal and shear tractions that act on planes whose outward unit normal vectors, in the principal coordinate system, are given by

$$(n_1, n_2, n_3) = (\pm 1/\sqrt{3}, \pm 1/\sqrt{3}, \pm 1/\sqrt{3}). \quad (2.147)$$

These planes are equally inclined to each of the three principal directions. There are eight such planes, each associated with a different set of choices for the signs appearing in (2.147). These eight planes are each parallel to a side of an octahedron whose vertices are located on the principal axes (Nadai, 1950, p. 105). From (2.86), the normal traction acting on any octahedral plane is

$$\sigma_{\text{oct}} = \sigma_1 n_1^2 + \sigma_2 n_2^2 + \sigma_3 n_3^2 = \frac{1}{3} (\sigma_1 + \sigma_2 + \sigma_3) = \frac{1}{3} I_1. \quad (2.148)$$

By (2.88), the magnitude of the shear traction acting on an octahedral plane is found to be given by

$$|\tau_{\text{oct}}| = \frac{1}{3} \{ (\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2 \}^{1/2} = \frac{\sqrt{2}}{3} \{ I_1^2 + 3I_2 \}^{1/2}. \quad (2.149)$$

The octahedral shear stress has the interesting physical interpretation of being equal to $\sqrt{5/3}$ times the root-mean-square shear stress, with all planes weighted equally.

Shear and normal stresses have different physical consequences, as the former act tangentially to a plane, and the latter act normal to it. The stress tensor, when written in a given coordinate system, explicitly contains only those shear stress components that act on planes whose normals are perpendicular to one of the three coordinate directions. Hence, when written in the principal coordinate system, the stress tensor seems to contain no shear stresses. However, in general it would be wrong to assume that no shear stresses are acting at that point, since the shears may be nonzero on oblique planes. Only in the special case in which all three principal stresses are *equal* is it true that the shear stresses on all planes are zero. This is most easily seen by referring to Mohr's circle, Fig. 2.9b, since in this case all three circles shrink down to a single point on the σ -axis.

It would be useful to have a way of representing the stress tensor that clearly showed whether or not there are *any* shear stresses acting at the point in question. To do this we decompose the stress tensor into an *isotropic* (or *hydrostatic*) part and a *deviatoric* part. The isotropic part of the stress is defined as

$$\tau^{\text{iso}} = \frac{1}{3} I_1 \mathbf{I} \equiv \tau_m \mathbf{I}, \quad (2.150)$$

where $\mathbf{I}$ is the 3×3 identity tensor, and τ_m is the mean value of the three principal stresses, or the *mean normal stress*. The mean normal stress is important in thermodynamic treatments of material deformation (McLellan, 1980), where it is analogous to the pressure that acts in a fluid. The deviatoric stress is obtained by subtracting the isotropic part of the stress tensor from the full stress tensor:

$$\mathbf{s} \equiv \boldsymbol{\tau}^{\text{dev}} = \boldsymbol{\tau} - \tau^{\text{iso}} = \boldsymbol{\tau} - \tau_m \mathbf{I}. \quad (2.151)$$

These equations can be written out explicitly as

$$\mathbf{s} = \begin{bmatrix} \tau_{xx} - \tau_m & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \tau_{yy} - \tau_m & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \tau_{zz} - \tau_m \end{bmatrix}, \quad (2.152)$$

$$\tau_m = \frac{1}{3} (\tau_{xx} + \tau_{yy} + \tau_{zz}). \quad (2.153)$$

The principal stress deviations are found from solving the eigenvalue problem $\mathbf{s}\mathbf{n} = s\mathbf{n}$, or $(\mathbf{s} - s\mathbf{I})\mathbf{n} = \mathbf{0}$, as in §2.5. But since $\mathbf{s} = \boldsymbol{\tau} - \tau_m \mathbf{I}$, the eigenvalue problem for the principal stress deviators is identical to that for the principal stresses themselves, except that the eigenvalue is now $s + \tau_m$ instead of σ . Hence,

the principal directions of the deviatoric stress are the same as the principal stress directions, and the principal deviatoric stresses are

$$s_1 = \sigma_1 - \tau_m = (2\sigma_1 - \sigma_2 - \sigma_3)/3, \quad (2.154)$$

$$s_2 = \sigma_2 - \tau_m = (2\sigma_2 - \sigma_1 - \sigma_3)/3, \quad (2.155)$$

$$s_3 = \sigma_3 - \tau_m = (2\sigma_3 - \sigma_1 - \sigma_2)/3. \quad (2.156)$$

The usefulness of this decomposition arises from the fact, (5.7), that in the elastic range of deformation, the mean stress τ_m controls the volumetric change of a rock, whereas the deviatoric stress $\mathbf{s}$ controls the distortion. Moreover, many of the criteria for failure are concerned primarily with distortion, in which case these criteria are most conveniently expressed in terms of the invariants of the stress deviation. In an arbitrary coordinate system, these invariants take the form

$$J_1 = s_{xx} + s_{yy} + s_{zz} = 0, \quad (2.157)$$

$$J_2 = s_{xy}^2 + s_{xz}^2 + s_{yz}^2 - s_{xx}s_{yy} - s_{xx}s_{zz} - s_{yy}s_{zz}, \quad (2.158)$$

$$J_3 = s_{xx}s_{yy}s_{zz} + 2s_{xy}s_{xz}s_{yz} - s_{xx}s_{yz}^2 - s_{yy}s_{xz}^2 - s_{zz}s_{xy}^2. \quad (2.159)$$

The first invariant of the deviatoric stress is always identically zero. The other two can be written in terms of the principal deviatoric stresses as follows:

$$J_2 = -(s_1s_2 + s_1s_3 + s_2s_3), \quad (2.160)$$

$$J_3 = s_1s_2s_3. \quad (2.161)$$

Algebraic manipulation of the previous equations leads to the following alternative forms for J_2 , which is the invariant that appears most often in failure criteria (Westergaard, 1952, p. 70):

$$J_2 = \frac{1}{2}(\tau_{xx}^2 + \tau_{yy}^2 + \tau_{zz}^2) + \tau_{xy}^2 + \tau_{yz}^2 + \tau_{xz}^2, \quad (2.162)$$

$$= \frac{1}{6}[(\tau_{xx} - \tau_{yy})^2 + (\tau_{xx} - \tau_{zz})^2 + (\tau_{yy} - \tau_{zz})^2] + \tau_{xy}^2 + \tau_{yz}^2 + \tau_{xz}^2, \quad (2.163)$$

$$= \frac{1}{6}[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2], \quad (2.164)$$

$$= \frac{1}{2}(s_1^2 + s_2^2 + s_3^2), \quad (2.165)$$

$$= 3(\tau_m)^2 + I_2, \quad (2.166)$$

$$= \frac{3}{2}(\tau_{\text{oct}})^2. \quad (2.167)$$

2.9 Displacement and strain

In rock mechanics, as in the mechanics of particles and rigid bodies, the fundamental kinematic variable is the *displacement*, which is the vector that quantifies the change in the position of a given particle of rock. In the typical rock mechanics problem, the position of each rock particle is labeled by its location, relative to

some coordinate system, in some state that is taken to be the “initial” state of the rock. This position can be denoted by $\mathbf{x} = (x, y, z)$. Loads are then applied to the rock, causing the rock particle that was initially located at point $\mathbf{x}$ to be displaced to a new position, $\mathbf{x}^* = (x^*, y^*, z^*)$. The vector that connects the original position $\mathbf{x}$ and the final position $\mathbf{x}^*$ is known as the “displacement of the particle that was initially at point $\mathbf{x}$,” or simply the “displacement at $\mathbf{x}$.” This vector is denoted in vector notation by $\mathbf{u}$, and its components are (u, v, w) . To be consistent with the sign convention used for tractions, in which a traction component is represented by a positive number if it points in the *negative* coordinate direction, the displacement vector must be defined according to

$$\mathbf{x}^* = \mathbf{x} - \mathbf{u}, \quad \text{that is,} \quad x^* = x - u, \quad y^* = y - v, \quad z^* = z - w. \quad (2.168)$$

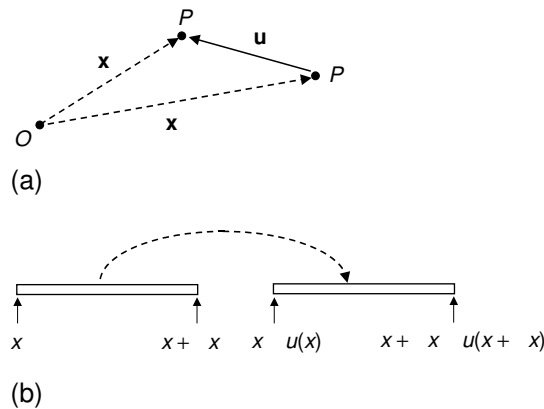
The displacement $\mathbf{u}$ can be interpreted as a vector that points *from* the new position, $\mathbf{x}^*$, *toward* the original position, $\mathbf{x}$ (Fig. 2.11a). In general, the displacement will vary from point to point, so that each component (u, v, w) will vary with all three position coordinates, x, y , and z .

The objective in solving a rock mechanics problem is to calculate the displacement vector $\mathbf{u}$ at every point in the rock mass or rock specimen, based on knowledge of the applied surface tractions and body forces, and the boundary conditions. To do this, it is necessary to introduce a set of intermediate quantities known as the *strains*. The reason, as will be seen in §5.2, is that the stresses are more directly related to the strains than to the displacements themselves. The use of strain rather than displacement is somewhat analogous to the use of stress, rather than the more physically obvious variable, the traction. Consequently, much of the theory of solid mechanics, and in particular elasticity, deals with stresses and strains, rather than tractions (forces) and displacements. However, in contrast to many other areas of solid mechanics, in which the displacements are often not of much intrinsic interest, in rock mechanics the displacement itself is often extremely important. Examples of such situations include mine closure, wellbore breakouts, and surface subsidence above mines and reservoirs.

Strain is essentially a measure of the relative displacement of nearby particles, rather than a measure of their absolute displacement. The basic concept behind

Fig. 2.11

(a) Displacement vector $\mathbf{u}$ of the piece of rock that is initially located at $\mathbf{x}$, (b) Generic displacement of a one-dimensional bar, used to define the normal strain.



the strain can be introduced in a one-dimensional context (Fig. 2.11b). Consider a short one-dimensional bar, initially of length L , whose left edge is initially located at point x , and whose right edge is located at point $x + \Delta x$. The initial length of this bar is given by $L = \Delta x$. This bar is now assumed to be deformed, such that the left edge of the bar moves to the location $x - u(x)$, and the right edge of the bar moves to the position $[x + \Delta x] - u(x + \Delta x)$. The new length of the bar is therefore equal to $L^* = \{[x + \Delta x] - u(x + \Delta x)\} - [x - u(x)]$. We now define the mean strain undergone by this bar as the *fractional decrease in the length of the bar*, that is,

$$\varepsilon = \frac{L - L^*}{L} = \frac{\Delta x - \{[x + \Delta x] - u(x + \Delta x) - [x - u(x)]\}}{\Delta x} = \frac{u(x + \Delta x) - u(x)}{\Delta x}. \quad (2.169)$$

According to this definition, the strain will be a positive number if the bar becomes shorter, and vice versa. Hence, positive strains represent contractions, and negative strains represent extensions. The strain *at the point x* is found by taking the limit of an infinitesimally short bar, which is mathematically equivalent to letting the initial length of the bar go to zero:

$$\varepsilon(x) = \lim_{L \rightarrow 0} \frac{L - L^*}{L} = \lim_{\Delta x \rightarrow 0} \frac{u(x + \Delta x) - u(x)}{\Delta x} \equiv \frac{du}{dx}. \quad (2.170)$$

The strain is therefore seen to be related to the spatial derivative of the displacement.

The type of strain described above, which is called a *normal strain*, can be generalized in an obvious way to two or three dimensions. However, in higher dimensions there are other types of strains, called the *shear strains*, which measure angular distortion, rather than stretching. The more general case of two- or three-dimensional strain is discussed in the following sections.

2.10 Infinitesimal strain in two dimensions

Consider a particle that is initially located at point $P = (x, y)$, as in Fig. 2.12a. Now consider a second particle that is initially located at $Q = (x + \Delta x, y)$, and a third particle located at $R = (x, y + \Delta y)$. The rock is then assumed to be deformed, such that these three particles move to positions P^* , Q^* , and R^* . The coordinates of these new locations are

$$\begin{aligned} P^* &= P - \mathbf{u}(P) = (x, y) - [u(x, y), v(x, y)] \\ &= [x - u(x, y), y - v(x, y)], \end{aligned} \quad (2.171)$$

$$\begin{aligned} Q^* &= Q - \mathbf{u}(Q) = (x + \Delta x, y) - [u(x + \Delta x, y), v(x + \Delta x, y)] \\ &= [x + \Delta x - u(x + \Delta x, y), y - v(x + \Delta x, y)], \end{aligned} \quad (2.172)$$

$$\begin{aligned} R^* &= R - \mathbf{u}(R) = (x, y + \Delta y) - [u(x, y + \Delta y), v(x, y + \Delta y)] \\ &= [x - u(x, y + \Delta y), y + \Delta y - v(x, y + \Delta y)]. \end{aligned} \quad (2.173)$$

Also shown in Fig. 2.12a are the points Q'' , which is the projection of point Q^* onto the x -axis that passes through point P^* , and R'' , which is the projection

of point R^* onto the y -axis that passes through point P^* . For example, Q'' will have the same x -component as Q^* and the same y -component as P^* and likewise for R'' .

We now express each of the displacements that appear in (2.171)–(2.173) as Taylor series taken about the point (x, y) . As the increments Δx and Δy are infinitesimally small, all terms in the Taylor series higher than the first-order terms, which are linear in Δx and Δy , can be ignored. For example, $u(x, y + \Delta y) = u(x, y) + (\partial u / \partial y) \Delta y$, etc., where the partial derivative is understood to be evaluated at (x, y) . The positions of the five points shown in Fig. 2.12a can therefore be expressed as

$$P^* = (x - u, y - v), \quad (2.174)$$

$$Q^* = (x + \Delta x - u - \frac{\partial u}{\partial x} \Delta x, y - v - \frac{\partial v}{\partial x} \Delta x), \quad (2.175)$$

$$R^* = (x - u - \frac{\partial u}{\partial y} \Delta y, y + \Delta y - v - \frac{\partial v}{\partial y} \Delta y) \quad (2.176)$$

$$Q'' = (x + \Delta x - u - \frac{\partial u}{\partial x} \Delta x, y - v), \quad (2.177)$$

$$R'' = (x - u, y + \Delta y - v - \frac{\partial v}{\partial y} \Delta y), \quad (2.178)$$

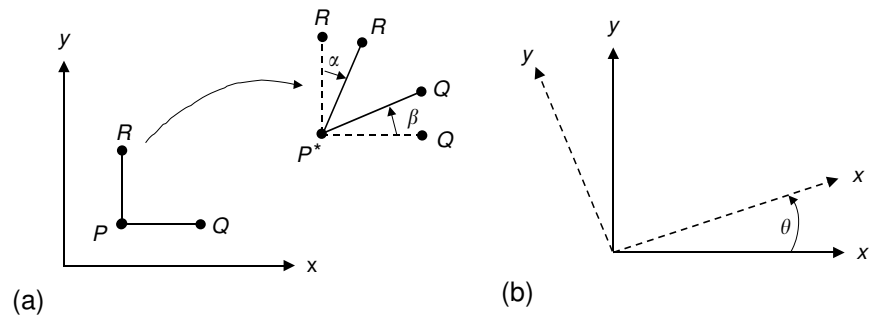
where it is understood that u , v , and their partial derivatives are evaluated at the point (x, y) .

The normal strain in the x direction, which is denoted by ε_{xx} , is now defined, as in §2.9, as the fractional shortening of a line element that is initially oriented along the x -axis. In other words, the strain ε_{xx} at point (x, y) is equal to the fractional contraction undergone by the element PQ , in the limit as $\Delta x \rightarrow 0$. Initially, the length of element PQ , which we denote by $|PQ|$, is Δx . After deformation, the length $|P^*Q^*|$ is found using the Pythagorean theorem:

$$\begin{aligned} |P^*Q^*|^2 &= |P^*Q''|^2 + |Q^*Q''|^2 = \left(\Delta x - \frac{\partial u}{\partial x} \Delta x \right)^2 + \left(\frac{\partial v}{\partial x} \Delta x \right)^2 \\ &= (\Delta x)^2 \left[1 - 2 \frac{\partial u}{\partial x} + \left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial x} \right)^2 \right]. \end{aligned} \quad (2.179)$$

Fig. 2.12

(a) Displacement of two small line segments PQ and PR that are initially at right angles to each other. (b) New (primed) coordinate system rotated by angle θ from original (unprimed) coordinate system.



The crucial assumption is now made that all partial derivatives of the displacements are much smaller than unity, which is equivalent to restricting the displacements not to vary too abruptly from point to point. This assumption leads to the theory of *infinitesimal strain*, within which theory the strains are linear functions of the partial derivatives of the displacement components. This linearity is a tremendous aid in solving specific problems. Moreover, the strains that usually occur in situations of engineering interest, such as around boreholes, tunnels, mines, etc., as well as during processes such as seismic wave propagation, are in fact often very small. For these reasons, the theory of infinitesimal strain is almost universally used in rock mechanics analyses. The theory of finite strain, which is much more complicated, is nevertheless needed for some geological processes in which the small-strain assumption is not valid; this theory is briefly discussed in §2.15.

Under the assumption that the partial derivatives of the displacement are small, we neglect the squares of these derivatives in (2.179), and use the approximation that $(1 - 2\delta)^{1/2} \approx 1 - \delta$ when $\delta \ll 1$, to find

$$|P^*Q^*| = \Delta x \left(1 - \frac{\partial u}{\partial x} \right). \quad (2.180)$$

Hence, the normal strain in the x direction at the point (x, y) is given by

$$\varepsilon_{xx} = \lim_{\Delta x \rightarrow 0} \frac{|PQ| - |P^*Q^*|}{|PQ|} = \lim_{\Delta x \rightarrow 0} \frac{\Delta x - \Delta x(1 - \partial u/\partial x)}{\Delta x} = \frac{\partial u}{\partial x}. \quad (2.181)$$

A similar analysis shows that the normal strain in the y direction, ε_{yy} , is given by

$$\varepsilon_{yy} = \lim_{\Delta y \rightarrow 0} \frac{|PR| - |P^*R^*|}{|PR|} = \lim_{\Delta y \rightarrow 0} \frac{\Delta y - \Delta y(1 - \partial v/\partial y)}{\Delta y} = \frac{\partial v}{\partial y}. \quad (2.182)$$

There are other types of distortion, other than stretching or contraction, which can also be identified and quantified. Figure 2.12a shows that the three points $\{P, Q, R\}$ initially form a right angle, but, in the case shown, form an acute angle after deformation. The change in this angle is known as the *shear strain*. Specifically, the shear strain ε_{xy} is defined as one-half of the *increase* in the angle initially formed by two infinitesimal line segments that initially lie parallel to the x and y axes, that is,

$$\varepsilon_{xy} = \frac{1}{2} \lim_{\Delta x, \Delta y \rightarrow 0} (\angle R^*P^*Q^* - \angle RPQ). \quad (2.183)$$

From Fig. 2.12a we see that $\angle R^*P^*Q^* = 90^\circ - \alpha - \beta$, whereas $\angle RPQ = 90^\circ$, by construction. The angle α is calculated from

$$\tan \alpha = \frac{|R^*R''|}{|P^*R''|} = \frac{-(\partial u/\partial y)\Delta y}{\Delta y(1 - (\partial v/\partial y))} \approx -\left(\frac{\partial u}{\partial y}\right), \quad (2.184)$$

where, due to the smallness of the partial derivatives, we have ignored the term $\partial v/\partial y$ in the denominator. But the smallness of the partial derivatives also allows

us to approximate the angle α by $\tan \alpha$, so that $\alpha = -(\partial u / \partial y)$. Similarly, it can be shown that $\beta = -(\partial v / \partial x)$. Combining these results with (2.183) leads to

$$\varepsilon_{xy} = \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right). \quad (2.185)$$

The pattern embodied in (2.185) is that the shear strain ε_{xy} is equal to the mean of the partial derivative of the displacement in the x direction with respect to y and the partial derivative of the displacement in the y direction with respect to x . By this definition, we see that if a shear strain ε_{yx} is defined, it will necessarily be equal to ε_{xy} . These four strains can be thought of as the four components of the *strain matrix*, $\boldsymbol{\varepsilon}$:

$$\boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_{xx} & \varepsilon_{xy} \\ \varepsilon_{yx} & \varepsilon_{yy} \end{bmatrix} = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) \\ \frac{1}{2} \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) & \frac{\partial v}{\partial y} \end{bmatrix}. \quad (2.186)$$

The shear strains are sometimes denoted by Γ_{xy} and Γ_{yx} . We will use the same basic symbol, ε , for all the strains, in order to emphasize that they are all components of the strain matrix. The shear strains will be distinguished by being the off-diagonal terms of this matrix, and therefore have mixed subscripts, whereas the normal strains are diagonal components, and have repeated subscripts. Another notation that had in the past been widely used, but is becoming obsolete, is to define “engineering” shear strains by $\gamma_{xy} = 2\varepsilon_{xy} = 2\Gamma_{xy}$. This definition seems to offer few advantages, and has the disadvantage that the matrix formed by the normal strains and the engineering shear strains does not constitute a tensor (see below).

The strain matrix is equal to the symmetric part of a matrix that is known as the *displacement gradient*, $\nabla \mathbf{u}$, whose components are the partial derivatives of the displacements with respect to the two coordinates, that is,

$$\boldsymbol{\varepsilon} = \text{sym}(\nabla \mathbf{u}) \equiv \frac{1}{2} [\nabla \mathbf{u} + (\nabla \mathbf{u})^T], \quad (2.187)$$

where

$$\nabla \mathbf{u} = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{bmatrix}. \quad (2.188)$$

The displacement gradient, and hence also the strain, is a second-order tensor, in the sense that if the coordinate system is rotated by an angle θ , the components of the strain in the new coordinate system are given by equations of the same form as (2.25)–(2.27). To prove this, consider a rotation of the coordinate system (Fig. 2.12b) such that the new x' -axis is rotated by counterclockwise angle θ from the original x -axis, that is,

$$x' = x \cos \theta + y \sin \theta, \quad y' = -x \sin \theta + y \cos \theta. \quad (2.189)$$

The inverse transformation, which is needed in the following calculations, is

$$x = x' \cos \theta - y' \sin \theta, \quad y = x' \sin \theta + y' \cos \theta. \quad (2.190)$$

As the displacement is also a vector, its components transform according to the same law:

$$u' = u \cos \theta + v \sin \theta, \quad v' = -u \sin \theta + v \cos \theta. \quad (2.191)$$

Now consider a component of the strain in the new coordinate system, such as, for example, $\varepsilon_{x'x'}$. Using the chain rule,

$$\begin{aligned} \varepsilon_{x'x'} &= \frac{\partial u'}{\partial x'} = \frac{\partial u'}{\partial u} \frac{\partial u}{\partial x} \frac{\partial x}{\partial x'} + \frac{\partial u'}{\partial u} \frac{\partial u}{\partial y} \frac{\partial y}{\partial x'} + \frac{\partial u'}{\partial v} \frac{\partial v}{\partial x} \frac{\partial x}{\partial x'} + \frac{\partial u'}{\partial v} \frac{\partial v}{\partial y} \frac{\partial y}{\partial x'} \\ &= \frac{\partial u}{\partial x} \cos^2 \theta + \frac{\partial u}{\partial y} \sin \theta \cos \theta + \frac{\partial v}{\partial x} \sin \theta \cos \theta + \frac{\partial v}{\partial y} \sin^2 \theta. \\ &= \varepsilon_{xx} \cos^2 \theta + 2\varepsilon_{xy} \sin \theta \cos \theta + \varepsilon_{yy} \sin^2 \theta. \end{aligned} \quad (2.192)$$

This is identical to the transformation law in two dimensions for τ_{xx} , (2.25). Similar analysis of the other components of the strain verifies that the strain in the primed coordinate system is related to that for the unprimed coordinate system by the same relationship that governs stress transformations, (2.30), that is,

$$\boldsymbol{\varepsilon}' = \mathbf{L} \boldsymbol{\varepsilon} \mathbf{L}^T, \quad (2.193)$$

where the rotation matrix $\mathbf{L}$ is given by (2.28). This proves that the strain is a second-order tensor.

The strain is also *symmetric*, by (2.187). In consequence of the fact that strain is a symmetric second-order tensor, all theorems pertaining to principal stresses, principal stress directions, maximum shear stresses, etc., carry over directly to strains. For example, in two dimensions there will always be two mutually orthogonal directions of principal normal strain, such that the normal strain in one of these directions is the maximum normal strain and the other a minimum. In particular, the directions of principal normal strains are defined by

$$\tan 2\theta = \frac{2\varepsilon_{xy}}{\varepsilon_{xx} - \varepsilon_{yy}}, \quad (2.194)$$

and the two principal strains, ε_1 and ε_2 , are given by

$$\varepsilon_1, \varepsilon_2 = \frac{1}{2}(\varepsilon_{xx} + \varepsilon_{yy}) \pm \frac{1}{4}[(\varepsilon_{xy})^2 + (\varepsilon_{xx} - \varepsilon_{yy})^2]. \quad (2.195)$$

Relative to the principal axes, the normal and shear strains in some direction that is rotated by an angle θ from the direction of the maximum normal strain, which in this context will be denoted by ε and Γ , are given by

$$\varepsilon = \varepsilon_1 \cos^2 \theta + \varepsilon_2 \sin^2 \theta, \quad (2.196)$$

$$\Gamma = -\frac{1}{2}(\varepsilon_1 - \varepsilon_2) \sin 2\theta. \quad (2.197)$$

The Mohr representation in the (ε, Γ) plane and the various constructions based on it, follow precisely as in §2.4.

The strain tensor quantifies the stretching and distortion undergone by the rock. There are also types of deformation that do not lead to any strain. One obvious type of such strain-free deformation is a *rigid-body translation*, in which all particles of the rock are displaced by the same amount, that is, $\mathbf{x}^* = \mathbf{x} - \mathbf{a}$, where the displacement vector $\mathbf{a}$ does not vary from point to point. In component form, a two-dimensional rigid-body translation is described by

$$x^* = x - a, \quad y^* = y - b, \quad (2.198)$$

where (a, b) , the components of the vector $\mathbf{a}$, are constants. For this type of motion, $\nabla \mathbf{u} = \mathbf{0}$, and the strain is consequently zero.

It is clear from (2.187) that any deformation whose displacement gradient is antisymmetric will also lead to no infinitesimal strain. Such types of deformations are *infinitesimal rigid-body rotations*. To verify this, consider a rigid rotation of the rock in the counterclockwise direction, by some small angle φ . This is mathematically similar, but physically different, from the *rotation of the coordinate system* that was discussed above. This rigid-body rotation is described by

$$x^* = x \cos \varphi - y \sin \varphi, \quad y^* = x \sin \varphi + y \cos \varphi. \quad (2.199)$$

The coordinates (x^*, y^*) are the coordinates of the particle that was originally located at (x, y) , *after* the rock has been rotated, but still referred to the *original* coordinate system. Since $\mathbf{x}^* = \mathbf{x} - \mathbf{u}$, the displacement components are given by

$$u = x - x^* = x(1 - \cos \varphi) + y \sin \varphi, \quad (2.200)$$

$$v = y - y^* = -x \sin \varphi + y(1 - \cos \varphi). \quad (2.201)$$

The displacement gradient can be calculated as

$$\nabla \mathbf{u} = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{bmatrix} = \begin{bmatrix} 1 - \cos \varphi & \sin \varphi \\ -\sin \varphi & 1 - \cos \varphi \end{bmatrix}. \quad (2.202)$$

As the angle of rotation is small, we can expand out the trigonometric terms in Taylor series, and ignore all terms that are higher than first-order in φ , yielding

$$\nabla \mathbf{u} = \begin{bmatrix} 0 & \varphi \\ -\varphi & 0 \end{bmatrix}. \quad (2.203)$$

The displacement gradient corresponding to an infinitesimal rigid-body rotation is therefore antisymmetric, in the sense that $(\nabla \mathbf{u})^T = -(\nabla \mathbf{u})$. This type of rotation leads to no strain, since $2\boldsymbol{\varepsilon} = [(\nabla \mathbf{u}) + (\nabla \mathbf{u})^T] = \mathbf{0}$. In general, we can define the rotation tensor $\boldsymbol{\omega}$ as the antisymmetric part of the displacement gradient, that is,

$$\boldsymbol{\omega} = \text{asym}(\nabla \mathbf{u}) \equiv \frac{1}{2} [\nabla \mathbf{u} - (\nabla \mathbf{u})^T] = \begin{bmatrix} 0 & \frac{1}{2} \left(\frac{\partial u}{\partial y} - \frac{\partial v}{\partial x} \right) \\ -\frac{1}{2} \left(\frac{\partial u}{\partial y} - \frac{\partial v}{\partial x} \right) & 0 \end{bmatrix}, \quad (2.204)$$

in which case it follows from (2.187) and (2.204) that $\nabla \mathbf{u} = \boldsymbol{\varepsilon} + \boldsymbol{\omega}$.

The discussion just given shows that infinitesimal deformations can always be thought of as being composed of three additive components: (i) rigid-body translations, which give rise to no strain or rotation, (ii) stretching and/or distorting deformations, which give rise to a nonzero strain tensor, and (iii) infinitesimal rotations, which give rise to a nonzero rotation tensor. Because of the linear relationships between $\nabla \mathbf{u}$, $\boldsymbol{\epsilon}$, and $\boldsymbol{\omega}$, the infinitesimal strain and infinitesimal rotation tensors are each additive in a manner that obeys the commutative law. In other words, the strain that arises as a result of two sequential displacements, $\mathbf{u}^1$ and $\mathbf{u}^2$, is equal to the sum of the two strains, $\boldsymbol{\epsilon}^1$ and $\boldsymbol{\epsilon}^2$, taken in either order. A similar superposition law also applies to the rotation tensor.

2.11 Infinitesimal strain in three dimensions

The theory of infinitesimal strain in three dimensions is a straightforward generalization of the two-dimensional theory. The initial location of a particle P will be denoted by $P = (x, y, z)$, relative to some Cartesian coordinate system. After the rock is deformed, the new location of that particle will be $P^* = (x^*, y^*, z^*)$, as in Fig. 2.12a. The vector that points *from* the new position *to* the old position is the displacement vector $\mathbf{u}$, that is,

$$\mathbf{u}(P) = P - P^*, \quad \text{that is, } (u, v, w) = (x, y, z) - (x^*, y^*, z^*). \quad (2.205)$$

In most branches of mechanics, it is customary to define $\mathbf{u}(P) = P^* - P$. The change in sign used in rock mechanics is needed so as to be consistent with the “compression = positive” sign convention used for the stresses.

Following the definitions of the strain components given in §2.10, in three dimensions we define three normal strains as

$$\epsilon_{xx} = \frac{\partial u}{\partial x}; \quad \epsilon_{yy} = \frac{\partial v}{\partial y}; \quad \epsilon_{zz} = \frac{\partial w}{\partial z}. \quad (2.206)$$

The normal strain in the x direction, ϵ_{xx} , has the same interpretation as in two dimensions: it is the fractional shortening of an infinitesimal line element that is initially oriented along the x -axis; likewise for the other two normal strains. Merely defining the three-dimensional strains in the same manner as in two dimensions does not of course guarantee that these strains will have the same physical interpretation. However, it is easy to see that the three-dimensional version of the calculation given in (2.179)–(2.181) would lead only to an additional term of $(\partial w / \partial z)^2$ in the bracketed part of (2.179). Under the assumption of infinitesimal strain, this term would be dropped, leading again to (2.181).

The three-dimensional shear strains are also defined in analogy with the two-dimensional case:

$$\begin{aligned} \epsilon_{xy} = \epsilon_{yx} &= \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right), & \epsilon_{xz} = \epsilon_{zx} &= \frac{1}{2} \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right), \\ \epsilon_{yz} = \epsilon_{zy} &= \frac{1}{2} \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right). \end{aligned} \quad (2.207)$$

These strains have the same physical interpretation as in two dimensions. For example, the shear strain ϵ_{xy} represents the increase in the angle formed by two

infinitesimally small line segments that initially start at point (x, y, z) , and lie along the x and y directions, respectively. However, the proof that this is indeed the correct interpretation of the definition of ε_{xy} given in (2.207) requires more than a simple extension of the two-dimensional version, since these two line segments will not necessarily continue to lie in the x - y plane after deformation. We start with a three-dimensional generalization of (2.171)–(2.173):

$$P^* = P - \mathbf{u}(P) = (x, y, z) - [u(x, y, z), v(x, y, z), w(x, y, z)], \quad (2.208)$$

$$Q^* = Q - \mathbf{u}(Q) = (x + \Delta x, y, z) - [u(x + \Delta x, y, z), v(x + \Delta x, y, z), w(x + \Delta x, y, z)], \quad (2.209)$$

$$R^* = R - \mathbf{u}(R) = (x, y + \Delta y, z) - [u(x, y + \Delta y, z), v(x, y + \Delta y, z), w(x, y + \Delta y, z)]. \quad (2.210)$$

Expanding each displacement component in a Taylor series around the point (x, y, z) , and retaining only terms that are linear in the increments Δx and Δy , leads to

$$P^* = (x - u, y - v, z - w), \quad (2.211)$$

$$Q^* = (x + \Delta x - u - \frac{\partial u}{\partial x} \Delta x, y - v - \frac{\partial v}{\partial x} \Delta x, z - w - \frac{\partial w}{\partial x} \Delta x), \quad (2.212)$$

$$R^* = (x - u - \frac{\partial u}{\partial y} \Delta y, y + \Delta y - v - \frac{\partial v}{\partial y} \Delta y, z - w - \frac{\partial w}{\partial y} \Delta y), \quad (2.213)$$

where all of the partial derivatives are understood to be evaluated at the point $P = (x, y, z)$.

The angle formed by the line segments P^*Q^* and P^*R^* can be calculated from

$$(\vec{P^*Q^*}) \cdot (\vec{P^*R^*}) = |P^*Q^*| |P^*R^*| \cos(\angle R^*P^*Q^*), \quad (2.214)$$

where the superposed arrow indicates that the line segment must be treated as a *vector*. From (2.211) and (2.212), we have

$$P^*Q^* = \left(\Delta x - \frac{\partial u}{\partial x} \Delta x, -\frac{\partial v}{\partial x} \Delta x, -\frac{\partial w}{\partial x} \Delta x \right), \quad (2.215)$$

$$P^*R^* = \left(-\frac{\partial u}{\partial y} \Delta y, \Delta y - \frac{\partial v}{\partial y} \Delta y, -\frac{\partial w}{\partial y} \Delta y \right). \quad (2.216)$$

Again neglecting all products and higher powers of the displacement gradient components, we have

$$(\vec{P^*Q^*}) \cdot (\vec{P^*R^*}) = -\frac{\partial u}{\partial y} \Delta x \Delta y - \frac{\partial v}{\partial x} \Delta x \Delta y. \quad (2.217)$$

The lengths of the two segments P^*Q^* and P^*R^* are, from (2.180) and (2.182),

$$|P^*Q^*| = \Delta x \left(1 - \frac{\partial v}{\partial x} \right), \quad |P^*R^*| = \Delta y \left(1 - \frac{\partial v}{\partial y} \right). \quad (2.218)$$

Solving (2.214) for $\cos(\angle R^*P^*Q^*)$ yields, to first-order in the displacement gradient terms,

$$\begin{aligned}\cos(\angle R^*P^*Q^*) &= \frac{-((\partial u)/(\partial x) + (\partial v)/(\partial y)) \Delta x \Delta y}{\Delta x (1 - (\partial u)/(\partial x)) \Delta y (1 - (\partial v)/(\partial y))} \\ &= -\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right).\end{aligned}\quad (2.219)$$

But as $\angle R^*P^*Q^*$ is very close to 90° , we can say that

$$\cos(\angle R^*P^*Q^*) = -\sin(\angle R^*P^*Q^*) \approx -(\angle R^*P^*Q^*). \quad (2.220)$$

Recalling the fundamental definition of shear strain, (2.183), and the fact that $\angle RPQ = 90^\circ$, it follows from (2.219) and (2.220) that

$$\varepsilon_{xy} = \frac{1}{2} \lim_{\Delta x, \Delta y \rightarrow 0} (\angle R^*P^*Q^* - \angle RPQ) = \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right), \quad (2.221)$$

which verifies that the shear strains, as defined by (2.207), do indeed measure the angular distortions.

The nine strains defined in (2.206) and (2.207) are the components of the three-dimensional strain tensor, $\mathbf{\varepsilon}$:

$$\begin{aligned}\mathbf{\varepsilon} &= \begin{bmatrix} \varepsilon_{xx} & \varepsilon_{xy} & \varepsilon_{xz} \\ \varepsilon_{yx} & \varepsilon_{yy} & \varepsilon_{yz} \\ \varepsilon_{zx} & \varepsilon_{zy} & \varepsilon_{zz} \end{bmatrix} \\ &= \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) & \frac{1}{2} \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right) \\ \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) & \frac{\partial v}{\partial y} & \frac{1}{2} \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right) \\ \frac{1}{2} \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right) & \frac{1}{2} \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right) & \frac{\partial w}{\partial z} \end{bmatrix}.\end{aligned}\quad (2.222)$$

As in the two-dimensional case, the strain matrix is equal to the symmetric part of the *displacement gradient*, $\nabla \mathbf{u}$, whose components are the partial derivatives of the displacements with respect to the three coordinates, that is,

$$\mathbf{\varepsilon} = \text{sym}(\nabla \mathbf{u}) \equiv \frac{1}{2} [\nabla \mathbf{u} + (\nabla \mathbf{u})^T], \quad (2.223)$$

$$\nabla \mathbf{u} = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{bmatrix}. \quad (2.224)$$

The antisymmetric part of the displacement gradient is again defined to be the rotation tensor, ω , whose components are given by

$$\begin{aligned}\omega &= \text{asym}(\nabla \mathbf{u}) \equiv \frac{1}{2} [\nabla \mathbf{u} - (\nabla \mathbf{u})^T] = \begin{bmatrix} \omega_{xx} & \omega_{xy} & \omega_{xz} \\ \omega_{yx} & \omega_{yy} & \omega_{yz} \\ \omega_{zx} & \omega_{zy} & \omega_{zz} \end{bmatrix} \\ &= \begin{bmatrix} 0 & \frac{1}{2} \left(\frac{\partial u}{\partial y} - \frac{\partial v}{\partial x} \right) & \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) \\ -\frac{1}{2} \left(\frac{\partial u}{\partial y} - \frac{\partial v}{\partial x} \right) & 0 & \frac{1}{2} \left(\frac{\partial v}{\partial z} - \frac{\partial w}{\partial y} \right) \\ -\frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) & -\frac{1}{2} \left(\frac{\partial v}{\partial z} - \frac{\partial w}{\partial y} \right) & 0 \end{bmatrix}. \quad (2.225)\end{aligned}$$

Arguments exactly analogous to that given in (2.200)–(2.203) would show that, for example, ω_{xz} quantifies the infinitesimal rotation about the y -axis, etc. As there are only three independent components of the rotation tensor, these components are often identified as the elements of the *rotation vector*, $(\omega_x, \omega_y, \omega_z)$, where

$$\omega_x = \frac{1}{2} \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right), \quad \omega_y = \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right), \quad \omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right). \quad (2.226)$$

The particular choices of signs in these definitions are made such that, for example, ω_x will be positive if the rotation is counterclockwise when viewed *from* the positive x -axis, *toward* the origin (Chou and Pagano, 1992, p. 48). It is important to note that the strain and rotation components typically vary from point to point, so that the rotations we are speaking of actually are confined to an infinitesimal neighborhood of the point (x, y, z) .

The three-dimensional demonstration that the strain is indeed a second-order tensor follows exactly the same lines as given in (2.192) for the two-dimensional case, except that there would be nine terms on the right-hand side, rather than four. The explicit expressions for the transformation of the six strain components are identical in form to (2.102)–(2.107). Hence, as the three-dimensional strain is a symmetric second-order tensor, all of the general theorems developed for three-dimensional stresses carry over to the strains. In particular, there will always be three mutually perpendicular directions in which the normal strain takes on locally stationary values, typically a maximum, a minimum, and a saddle point. These three principal normal strains are denoted by $\varepsilon_1 \geq \varepsilon_2 \geq \varepsilon_3$. In the *principal coordinate system*, in which the three principal directions serve as the coordinate axes, the shear strains are zero, and the strain tensor is purely diagonal. (The question of whether or not the principal directions of strain coincide with the principal directions of stress depends on the stress–strain law of the rock; see Chapter 5). Hence, at each point, there is always one coordinate system in which the deformation of the rock can be represented by stretching (or contraction) along three mutually perpendicular directions. Of course, there are also,

in general, rigid-body translations and rotations occurring at each point. Therefore, the most general infinitesimal displacement at a point is a combination of stretching (by, in general, differing amounts) along three mutually perpendicular axes, followed by a rigid-body rotation and then a rigid-body translation. If the rotations are infinitesimal, it can be shown (Fung, 1965, p. 97) that subsequent rotations about the (x, y, z) axes by angles $(\omega_x, \omega_y, \omega_z)$ are equivalent to a single rotation about an axis that coincides with the rotation vector, the angle of which is given by $\omega = (\omega_x^2 + \omega_y^2 + \omega_z^2)^{1/2}$.

The values of the three principal strains are found by solving an equation analogous to (2.69):

$$\varepsilon^3 - I_1 \varepsilon^2 - I_2 \varepsilon - I_3 = 0, \quad (2.227)$$

in which the three *strain invariants* are defined by

$$I_1 = \text{trace}(\mathbf{\varepsilon}) = \varepsilon_{xx} + \varepsilon_{yy} + \varepsilon_{zz}, \quad (2.228)$$

$$I_2 = \frac{1}{2} \{ \text{trace}(\mathbf{\varepsilon}^2) - [\text{trace}(\mathbf{\varepsilon})]^2 \} = \varepsilon_{xy}^2 + \varepsilon_{xz}^2 + \varepsilon_{yz}^2 - \varepsilon_{xx}\varepsilon_{yy} - \varepsilon_{xx}\varepsilon_{zz} - \varepsilon_{yy}\varepsilon_{zz}, \quad (2.229)$$

$$I_3 = \det(\mathbf{\varepsilon}) = \varepsilon_{xx}\varepsilon_{yy}\varepsilon_{zz} + 2\varepsilon_{xy}\varepsilon_{xz}\varepsilon_{yz} - \varepsilon_{xx}\varepsilon_{yz}^2 - \varepsilon_{yy}\varepsilon_{xz}^2 - \varepsilon_{zz}\varepsilon_{xy}^2. \quad (2.230)$$

The first invariant of strain has the physical interpretation of representing the volumetric strain in the vicinity of the point (x, y, z) . To prove this, consider a small cube of rock, one of whose corners is initially located at (x, y, z) , and whose edges are aligned (before deformation) with the three principal directions of strain. Before the deformation, the edges of this cube each had length L , and the cube had volume V . After the deformation, the edges of this cube will have length $L(1 - \varepsilon_1)$, $L(1 - \varepsilon_2)$, and $L(1 - \varepsilon_3)$, and the new volume will be $V^* = L^3(1 - \varepsilon_1)(1 - \varepsilon_2)(1 - \varepsilon_3)$. The volumetric strain, defined as the fractional *decrease* in the volume of the cube, is

$$\begin{aligned} \varepsilon_v &= \lim_{L \rightarrow 0} \frac{V - V^*}{V} = \frac{L^3(1 - \varepsilon_1)(1 - \varepsilon_2)(1 - \varepsilon_3) - L^3}{L^3} \\ &= (1 - \varepsilon_1)(1 - \varepsilon_2)(1 - \varepsilon_3) - 1 \\ &= 1 - (\varepsilon_1 + \varepsilon_2 + \varepsilon_3) + (\varepsilon_1\varepsilon_2 + \varepsilon_1\varepsilon_3 + \varepsilon_2\varepsilon_3) - \varepsilon_1\varepsilon_2\varepsilon_3 - 1 \\ &\approx \varepsilon_1 + \varepsilon_2 + \varepsilon_3 = I_1, \end{aligned} \quad (2.231)$$

where, in the last step, the smallness of the strains has been invoked. Another commonly used symbol for the volumetric strain is Δ . As I_1 is an invariant, the volumetric strain can be calculated from (2.228), in *any* coordinate system.

The strain tensor can be decomposed into an isotropic and a deviatoric part, in a manner exactly analogous to the decomposition of the stresses:

$$\mathbf{\varepsilon} = \mathbf{\varepsilon}^{\text{iso}} + \mathbf{\varepsilon}^{\text{dev}}, \quad (2.232)$$

$$\mathbf{\varepsilon}^{\text{iso}} = \frac{1}{3} I_1 \mathbf{I} = \frac{1}{3} \varepsilon_v \mathbf{I}, \quad (2.233)$$

$$\mathbf{e} = \boldsymbol{\varepsilon}^{\text{dev}} = \boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^{\text{iso}} = \boldsymbol{\varepsilon} - \frac{1}{3}\varepsilon_v \mathbf{I}. \quad (2.234)$$

Equation (2.234) can be written out explicitly as

$$\mathbf{e} = \begin{bmatrix} \varepsilon_{xx} - \frac{1}{3}\varepsilon_v & \varepsilon_{xy} & \varepsilon_{xz} \\ \varepsilon_{yx} & \varepsilon_{yy} - \frac{1}{3}\varepsilon_v & \varepsilon_{yz} \\ \varepsilon_{zx} & \varepsilon_{zy} & \varepsilon_{zz} - \frac{1}{3}\varepsilon_v \end{bmatrix}. \quad (2.235)$$

The *mean normal strain* is defined by

$$\varepsilon_m = \frac{1}{3}I_1 = \frac{1}{3}\varepsilon_v. \quad (2.236)$$

The principal axes of the deviatoric strain coincide with the principal axes of strain, and the magnitudes of the principal deviatoric strains are

$$e_1 = \varepsilon_1 - \varepsilon_m, \quad e_2 = \varepsilon_2 - \varepsilon_m, \quad e_3 = \varepsilon_3 - \varepsilon_m, \quad (2.237)$$

so that $e_1 + e_2 + e_3 = 0$.

An important special type of deformation is a *homogeneous* deformation, in which the strains and rotations are the same at each point. In this case, it follows from (2.223)–(2.226) that the displacement gradient takes the form

$$\nabla \mathbf{u} = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{bmatrix} = \begin{bmatrix} \varepsilon_{xx}^0 & \varepsilon_{xy}^0 - \omega_z^0 & \varepsilon_{xz}^0 + \omega_y^0 \\ \varepsilon_{xy}^0 + \omega_z^0 & \varepsilon_{yy}^0 & \varepsilon_{yz}^0 - \omega_x^0 \\ \varepsilon_{xz}^0 - \omega_y^0 & \varepsilon_{yz}^0 + \omega_x^0 & \varepsilon_{zz}^0 \end{bmatrix}, \quad (2.238)$$

where the superscript 0 indicates that these components do not vary with position. The nine first-order partial differential equations represented by (2.238) can be integrated to yield

$$u = \varepsilon_{xx}^0 x + (\varepsilon_{xy}^0 - \omega_z^0)y + (\varepsilon_{xz}^0 + \omega_y^0)z + a, \quad (2.239)$$

$$v = (\varepsilon_{xy}^0 + \omega_z^0)x + \varepsilon_{yy}^0 y + (\varepsilon_{yz}^0 - \omega_x^0)z + b, \quad (2.240)$$

$$w = (\varepsilon_{xz}^0 - \omega_y^0)x + (\varepsilon_{yz}^0 + \omega_x^0)y + \varepsilon_{zz}^0 z + c, \quad (2.241)$$

where $\{a, b, c\}$ are constants. This deformation represents a stretching in the $\{x, y, z\}$ directions by fractional amounts $\{\varepsilon_{xx}^0, \varepsilon_{yy}^0, \varepsilon_{zz}^0\}$, followed by rotations about the $\{x, y, z\}$ axes by amounts $\{\omega_x^0, \omega_y^0, \omega_z^0\}$, finally followed by rigid-body translations in the $\{x, y, z\}$ directions by amounts $\{a, b, c\}$. It is easy to show that, under a homogeneous deformation: (i) a straight line remains straight, (ii) two parallel lines remain parallel, and (iii) spheres are transformed into ellipses. An interesting aspect of homogeneous deformations is that the above-listed properties hold *regardless* of whether or not the strain and rotation components are infinitesimal.

2.12 Determination of principal stresses or strains from measurements

Various methods exist to measure the state of stress underground. Many of these methods actually involve measurement of the strains, from which the stresses are estimated using known stress-strain relations for the given rock (see §5.2). These methods, which involve strain gauges or displacement meters, are capable of measuring normal strains in a given direction. The question then arises of using the measured normal strains to find the principal strains and the principal directions. As the problem is mathematically identical in the case of stresses or strains, we will discuss it in the context of finding the principal strains and their directions.

In two dimensions, there are three independent strain components, and so it seems plausible that three strain measurements will be needed to supply sufficient information to find these components. Another way of looking at this problem is that three pieces of data are needed to estimate the two principal strains and the angle of rotation of one of the principal directions with respect to some arbitrary direction. Assume that we have measured the normal strains in three directions, P , Q , and R , in some plane, as in Fig. 2.13. The angles of rotation between OP , OQ , and OR will be known, but the angle of rotation θ between, say, OP and the direction of the major principal stress will not be known a priori. In general, it will often not be the case that two principal stresses actually lie in the plane in question; in this case, we will actually be dealing with the *subsidiary principal stresses*, as defined in §2.6; this does not affect the following discussion.

From (2.54) we have

$$\varepsilon_{PP} = \frac{1}{2}(\varepsilon_1 + \varepsilon_2) + \frac{1}{2}(\varepsilon_1 - \varepsilon_2) \cos 2\theta, \quad (2.242)$$

$$\varepsilon_{QQ} = \frac{1}{2}(\varepsilon_1 + \varepsilon_2) + \frac{1}{2}(\varepsilon_1 - \varepsilon_2) \cos 2(\theta + \alpha), \quad (2.243)$$

$$\varepsilon_{RR} = \frac{1}{2}(\varepsilon_1 + \varepsilon_2) + \frac{1}{2}(\varepsilon_1 - \varepsilon_2) \cos 2(\theta + \alpha + \beta). \quad (2.244)$$

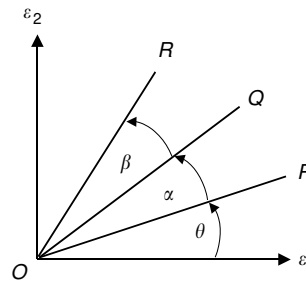
These equations can be solved for ε_1 , ε_2 , and θ . For the commonly used strain gauge rosette configuration in which $\alpha = \beta = 45^\circ$, the solution is

$$\varepsilon_1 + \varepsilon_2 = \varepsilon_{PP} + \varepsilon_{RR}, \quad (2.245)$$

$$\varepsilon_1 - \varepsilon_2 = [(\varepsilon_{PP} - 2\varepsilon_{QQ} + \varepsilon_{RR})^2 + (\varepsilon_{PP} - \varepsilon_{RR})^2]^{1/2}, \quad (2.246)$$

$$\theta = \frac{1}{2} \arctan[(\varepsilon_{PP} - 2\varepsilon_{QQ} + \varepsilon_{RR})/(\varepsilon_{PP} - \varepsilon_{RR})]. \quad (2.247)$$

Fig. 2.13 Normal strains that are measured in three directions, OP , OQ , and OR can be used to find the principal normal strains and their orientation, as explained in the text.



Another common case is $\alpha = \beta = 60^\circ$, for which the solution is

$$\varepsilon_1 + \varepsilon_2 = \frac{2}{3}(\varepsilon_{PP} + \varepsilon_{QQ} + \varepsilon_{RR}), \quad (2.248)$$

$$\varepsilon_1 - \varepsilon_2 = \left[\frac{4}{3}(\varepsilon_{QQ} - \varepsilon_{RR})^2 + \frac{4}{9}(2\varepsilon_{PP} - \varepsilon_{QQ} - \varepsilon_{RR})^2 \right]^{1/2}, \quad (2.249)$$

$$\theta = \frac{1}{2} \arctan[\sqrt{3}(\varepsilon_{QQ} - \varepsilon_{RR})/(\varepsilon_{QQ} + \varepsilon_{RR} - 2\varepsilon_{PP})]. \quad (2.250)$$

Solutions for other configurations are described by Hetényi (1950, p. 412). These equations could also be solved graphically, using Mohr's circle, as in §2.4.

2.13 Compatibility equations

In two dimensions, there are two displacement functions and four strain components. Two of these strains, the shear strains ε_{xy} and ε_{yx} , are equal to each other by definition. In three dimensions, there are essentially six strains and three displacements. If the displacement field $\mathbf{u}$ is known as a function of $\mathbf{x}$, the strains can be computed by simply taking partial derivatives. The converse problem, that of determining the displacements from the strains, requires a complicated integration procedure (Sokolnikoff, 1956, pp. 25–9). In those rock mechanics problems for which the displacements themselves are of interest, one would usually use a solution procedure that directly makes use of the displacements as the basic unknown variables. However, there are many situations in which only the stresses are desired but the displacements are not. For these problems, solution procedures exist that utilize the stresses as the basic unknowns but in which the displacements do not explicitly appear (see §5.5 and §5.7). Hence, in both types of situations the integration procedure itself is usually avoided. Nevertheless, when using a stress-based formulation of the elasticity equations, the number of equations will be less than the number of unknowns unless one considers an additional set of equations known as the *compatibility equations*. These equations are necessary conditions in order that the partial differential equations represented by (2.186) or (2.222) can be integrated to yield the displacements. These equations must be satisfied whether or not one attempts to perform the integration.

To understand the origin of the integrability condition, consider the simpler, but essentially equivalent, situation of a single function f of two variables, x and y . Assume that the two partial derivatives of f are given as specified functions, M and N , that is,

$$M(x, y) = \frac{\partial f}{\partial x}, \quad N(x, y) = \frac{\partial f}{\partial y}. \quad (2.251)$$

The two functions M and N cannot be specified arbitrarily, even if they are both assumed to be continuous, differentiable functions. This is because, as the mixed partial derivatives of f must be equal (Lang, 1973, p. 110), we must have

$$\frac{\partial M}{\partial y} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial N}{\partial x}. \quad (2.252)$$

Hence, M and N are not completely independent, but must satisfy the partial differential equation $\partial M/\partial y = \partial N/\partial x$.

Similar integrability conditions exist for the strains, in both two and three dimensions. The integrability condition is easy to derive in two dimensions, simply by taking two partial derivatives of each of the three strain components, as given by (2.181), (2.182), and (2.185), which leads to

$$2 \frac{\partial^2 \varepsilon_{xy}}{\partial x \partial y} = \frac{\partial^2 \varepsilon_{xx}}{\partial y^2} + \frac{\partial^2 \varepsilon_{yy}}{\partial x^2}. \quad (2.253)$$

In three dimensions, six such relations can be derived among the six strain components. (The number of compatibility equations is *not* simply equal to the number of strains less the number of displacements, as is sometimes implied). These relations, first derived in 1860 by the great French elastician Saint Venant, are, in addition to (2.253),

$$2 \frac{\partial^2 \varepsilon_{yz}}{\partial y \partial z} = \frac{\partial^2 \varepsilon_{yy}}{\partial z^2} + \frac{\partial^2 \varepsilon_{zz}}{\partial y^2}, \quad (2.254)$$

$$2 \frac{\partial^2 \varepsilon_{zx}}{\partial z \partial x} = \frac{\partial^2 \varepsilon_{zz}}{\partial x^2} + \frac{\partial^2 \varepsilon_{xx}}{\partial z^2}, \quad (2.255)$$

$$\frac{\partial^2 \varepsilon_{xx}}{\partial y \partial z} = \frac{\partial}{\partial x} \left(-\frac{\partial \varepsilon_{yz}}{\partial x} + \frac{\partial \varepsilon_{zx}}{\partial y} + \frac{\partial \varepsilon_{xy}}{\partial z} \right), \quad (2.256)$$

$$\frac{\partial^2 \varepsilon_{yy}}{\partial z \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial \varepsilon_{yz}}{\partial x} - \frac{\partial \varepsilon_{zx}}{\partial y} + \frac{\partial \varepsilon_{xy}}{\partial z} \right), \quad (2.257)$$

$$\frac{\partial^2 \varepsilon_{zz}}{\partial x \partial y} = \frac{\partial}{\partial z} \left(\frac{\partial \varepsilon_{yz}}{\partial x} + \frac{\partial \varepsilon_{zx}}{\partial y} - \frac{\partial \varepsilon_{xy}}{\partial z} \right). \quad (2.258)$$

These equations are necessary, and also sufficient, for the displacements to exist as continuous, single-valued functions (Fung, 1965, pp. 101–3). The importance of the compatibility equations in solving elasticity problems is illustrated in §5.7. Finally, it should be noted that the compatibility equations impose conditions only on the manner in which the strains vary from point to point. But there are no necessary conditions that must hold between the strain components at any one point; at any given point, the strain components ε_{xx} , ε_{yy} , and ε_{xy} can take on any three arbitrary values.

2.14 Stress and strain in polar and cylindrical coordinates

In many rock mechanics problems involving circular geometries, such as those involving tunnels, boreholes, or cylindrical rock cores used as test specimens, it is convenient to use polar coordinates (for two-dimensional problems) or cylindrical coordinates (for three-dimensional problems). We will first consider polar coordinates in a plane; cylindrical coordinates are created by simply adding a Cartesian axis that is perpendicular to the plane.

Consider a plane, with an (x, y) coordinate system attached to some origin O (Fig. 2.14a). Each point P , with coordinates (x, y) relative to the Cartesian coordinate system, can also be identified with two polar coordinates, r and θ .

The radial coordinate r measures the distance from P to the origin, whereas the angular coordinate θ measures the angle of counterclockwise rotation from the line segment OX to OP . These two sets of coordinates are related to each other by

$$x = r \cos \theta, \quad y = r \sin \theta, \quad (2.259)$$

$$r = (x^2 + y^2)^{1/2}, \quad \theta = \arctan(y/x). \quad (2.260)$$

At each point P , we can imagine a local pair of unit vectors, $\mathbf{e}_r$ and $\mathbf{e}_\theta$, pointing in the r and θ directions, respectively. An infinitesimal region of rock bounded by the arcs r , $r + \Delta r$, θ , and $\theta + \Delta \theta$, is shown in Fig. 2.14b.

At each point P , the two unit vectors $\{\mathbf{e}_r, \mathbf{e}_\theta\}$ can be thought of as defining a Cartesian coordinate system that is rotated by an angle θ from the $\{\mathbf{e}_x, \mathbf{e}_y\}$ system. Hence, the stresses in the polar coordinate system can be found from those in the Cartesian system through equations (2.25)–(2.27):

$$\tau_{rr} = \tau_{xx} \cos^2 \theta + 2\tau_{xy} \sin \theta \cos \theta + \tau_{yy} \sin^2 \theta, \quad (2.261)$$

$$\tau_{\theta\theta} = \tau_{xx} \sin^2 \theta - 2\tau_{xy} \sin \theta \cos \theta + \tau_{yy} \cos^2 \theta, \quad (2.262)$$

$$\tau_{r\theta} = (\tau_{yy} - \tau_{xx}) \sin \theta \cos \theta + \tau_{xy} (\cos^2 \theta - \sin^2 \theta). \quad (2.263)$$

At each point P , the stresses in polar coordinates constitute a symmetric second-order tensor.

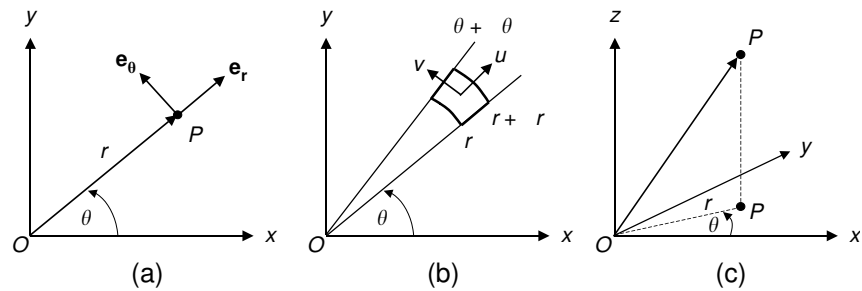
The displacement vector in polar coordinates (Fig. 2.14b) will be denoted by (u', v') . Note that, according to the sign convention introduced in (2.171), the actual motion of the rock particle is in the direction of $(-u', -v')$. Because the (r, θ) coordinate system is (locally) merely a rotated Cartesian coordinate system, the displacement components in the (r, θ) system are related to those in the (x, y) system by (2.191):

$$u' = u \cos \theta + v \sin \theta, \quad v' = -u \sin \theta + v \cos \theta, \quad (2.264)$$

$$u = u' \cos \theta - v' \sin \theta, \quad v = u' \sin \theta + v' \cos \theta. \quad (2.265)$$

The situation is more complicated for relations that involve differentiation, such as the strain–displacement relations. For such relationships, it becomes necessary to account for the fact that, in contrast to the unit vectors $\{\mathbf{e}_x, \mathbf{e}_y\}$, the unit vectors $\{\mathbf{e}_r, \mathbf{e}_\theta\}$ point in different directions at different locations. An implication of this fact is that it is not proper to define the strains in a polar

Fig. 2.14 (a) Polar coordinate system. (b) Infinitesimal element of rock in a polar coordinate system. (c) Cylindrical coordinate system.



coordinate system in the same manner as was done for Cartesian coordinates in §2.10. The proper definition of the strains in general, non-Cartesian coordinates is discussed by Green and Zerna (1954) and Amenzade (1979), among others. In polar coordinates, the proper definitions of strains can be arrived at by using the chain rule of partial differentiation, in conjunction with (2.259), (2.260), (2.264), and (2.265).

To carry out these calculations, we need the following partial derivatives:

$$\frac{\partial r}{\partial x} = \frac{\partial}{\partial x}(x^2 + y^2)^{1/2} = \frac{x}{(x^2 + y^2)^{1/2}} = \frac{x}{r} = \cos \theta, \quad (2.266)$$

$$\frac{\partial r}{\partial y} = \frac{\partial}{\partial y}(x^2 + y^2)^{1/2} = \frac{y}{(x^2 + y^2)^{1/2}} = \frac{y}{r} = \sin \theta, \quad (2.267)$$

$$\frac{\partial \theta}{\partial x} = \frac{\partial}{\partial x}[\arctan(y/x)] = \frac{-y/x^2}{[1 + (y/x)^2]} = \frac{-y}{r^2} = \frac{-\sin \theta}{r}, \quad (2.268)$$

$$\frac{\partial \theta}{\partial y} = \frac{\partial}{\partial y}[\arctan(y/x)] = \frac{1/x}{[1 + (y/x)^2]} = \frac{x}{r^2} = \frac{\cos \theta}{r}. \quad (2.269)$$

Now apply the chain rule to the strain component ε_{xx} :

$$\varepsilon_{xx} = \frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial u}{\partial \theta} \frac{\partial \theta}{\partial x} = \cos \theta \frac{\partial u}{\partial r} - \frac{\sin \theta}{r} \frac{\partial u}{\partial \theta}. \quad (2.270)$$

But, from (2.265),

$$\frac{\partial u}{\partial r} = \cos \theta \frac{\partial u'}{\partial r} - \sin \theta \frac{\partial v'}{\partial r}, \quad (2.271)$$

$$\frac{\partial u}{\partial \theta} = \cos \theta \frac{\partial u'}{\partial \theta} - u' \sin \theta - \sin \theta \frac{\partial v'}{\partial \theta} - v' \cos \theta. \quad (2.272)$$

Combining (2.270)–(2.272), and collecting terms, yields

$$\varepsilon_{xx} = \frac{\partial u'}{\partial r} \cos^2 \theta - \left[\frac{\partial v'}{\partial r} + \frac{1}{r} \frac{\partial u'}{\partial \theta} - \frac{v'}{r} \right] \sin \theta \cos \theta + \left[\frac{1}{r} \frac{\partial v'}{\partial \theta} + \frac{u'}{r} \right] \sin^2 \theta. \quad (2.273)$$

As the $\{\mathbf{e}_x, \mathbf{e}_y\}$ axes are obtained from the $\{\mathbf{e}_r, \mathbf{e}_\theta\}$ axes by a counterclockwise rotation of angle $-\theta$, the strain ε_{xx} can also be obtained from the strains in the polar coordinate system by the usual strain transformation equations, (2.192), using an angle of $-\theta$:

$$\varepsilon_{xx} = \varepsilon_{rr} \cos^2 \theta - 2\varepsilon_{r\theta} \sin \theta \cos \theta + \varepsilon_{\theta\theta} \sin^2 \theta. \quad (2.274)$$

Comparison of (2.273) and (2.274) indicates that

$$\varepsilon_{rr} = \frac{\partial u'}{\partial r}, \quad \varepsilon_{\theta\theta} = \frac{1}{r} \frac{\partial v'}{\partial \theta} + \frac{u'}{r}, \quad \varepsilon_{r\theta} = \frac{1}{2} \left(\frac{\partial v'}{\partial r} + \frac{1}{r} \frac{\partial u'}{\partial \theta} - \frac{v'}{r} \right). \quad (2.275)$$

These relations can also be derived by more sophisticated methods (Sokolnikoff, 1956, pp. 177–84).

The areal dilatation, which is the two-dimensional analogue of the volumetric strain, is given, as usual, by the sum of the diagonal components of the strain tensor:

$$\varepsilon_a = \varepsilon_{rr} + \varepsilon_{\theta\theta} = \frac{\partial u'}{\partial r} + \frac{u'}{r} + \frac{1}{r} \frac{\partial v'}{\partial \theta}, \quad (2.276)$$

The following compatibility criterion for the strains in polar coordinates can be derived by taking appropriate partial derivatives of the strains in (2.275):

$$2 \frac{\partial^2 (r \varepsilon_{r\theta})}{\partial r \partial \theta} = r \frac{\partial^2 (r \varepsilon_{\theta\theta})}{\partial r^2} - r \frac{\partial \varepsilon_{rr}}{\partial r} + \frac{\partial^2 \varepsilon_{rr}}{\partial \theta^2}. \quad (2.277)$$

In three-dimensional problems involving cylindrical geometry or axial symmetry, cylindrical polar coordinates are often used (Fig. 2.14c). The third unit vector $\mathbf{e}_z$ is perpendicular to the $\{\mathbf{e}_r, \mathbf{e}_\theta\}$ plane, and is oriented such that $\{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{e}_z\}$ form a right-handed coordinate system. The two coordinates r and θ are then taken to be the polar coordinates of the point P' , which is the projection of point P onto the (x, y) plane, and the coordinate z is the perpendicular distance of P from the (x, y) plane. The displacement component in the z direction will be denoted by w' , so that

$$\mathbf{u}' = u' \mathbf{e}_r + v' \mathbf{e}_\theta + w' \mathbf{e}_z. \quad (2.278)$$

The nine stress components are denoted by

$$\{\tau_{rr}, \tau_{r\theta}, \tau_{rz}, \tau_{\theta r}, \tau_{\theta\theta}, \tau_{\theta z}, \tau_{zr}, \tau_{z\theta}, \tau_{zz}\}, \quad (2.279)$$

where the subscripts play the usual role in indicating the direction of the traction and the plane on which it acts. The strain-displacement relations are given by (2.275), along with

$$\varepsilon_{zr} = \frac{1}{2} \left(\frac{\partial w'}{\partial r} + \frac{\partial u'}{\partial z} \right), \quad \varepsilon_{z\theta} = \frac{1}{2} \left(\frac{1}{r} \frac{\partial w'}{\partial \theta} + \frac{\partial v'}{\partial z} \right), \quad \varepsilon_{zz} = \frac{\partial w'}{\partial z}. \quad (2.280)$$

The volumetric strain is given by

$$\varepsilon_v = \varepsilon_{rr} + \varepsilon_{\theta\theta} + \varepsilon_{zz} = \frac{\partial u'}{\partial r} + \frac{u'}{r} + \frac{1}{r} \frac{\partial v'}{\partial \theta} + \frac{\partial w'}{\partial z}. \quad (2.281)$$

The five additional compatibility equations, which in the full three-dimensional case are needed in addition to (2.277), can be found in Rekach (1979, p. 27).

2.15 Finite strain

Most analyses in rock mechanics, including most of the commonly used computer codes, utilize the theory of infinitesimal strain. However, some problems, particularly those involving deformations that occur over geological time, necessarily involve strains that are not small compared to unity. To analyze these processes, the theory of finite strain must be used (Means, 1976; Oertel, 1996). This more general theory, which reduces to the theory of infinitesimal strain when the

strain and rotation components are small, is more difficult to utilize due mainly to the nonlinear relationships that exist between the components of the displacement gradient and the components of the finite strain tensor. This nonlinearity is essentially different from that which arises when using nonlinear stress–strain relationships in conjunction with the infinitesimal strain assumption; this latter type of nonlinearity is quite common in rock mechanics analyses.

Recall that the original location of a rock particle is denoted by $\mathbf{x} = (x, y, z)$, and its position after the rock has been deformed is denoted by $\mathbf{x}^* = (x^*, y^*, z^*)$. For the time being, we will not utilize the concept of displacement, which is not fundamental to the theory of finite strain. Now consider a small line element in the undeformed configuration, represented by the vector $d\mathbf{x}$, whose tail is located at $\mathbf{x}$. After the deformation, this element will be deformed into a new line element $d\mathbf{x}^*$, located at point $\mathbf{x}^*$, whose components are given by

$$dx^* = \frac{\partial x^*}{\partial x} dx + \frac{\partial x^*}{\partial y} dy + \frac{\partial x^*}{\partial z} dz, \quad (2.282)$$

$$dy^* = \frac{\partial y^*}{\partial x} dx + \frac{\partial y^*}{\partial y} dy + \frac{\partial y^*}{\partial z} dz, \quad (2.283)$$

$$dz^* = \frac{\partial z^*}{\partial x} dx + \frac{\partial z^*}{\partial y} dy + \frac{\partial z^*}{\partial z} dz. \quad (2.284)$$

This relationship between the deformed and undeformed elements can be written in vector-matrix form as

$$\begin{bmatrix} dx^* \\ dy^* \\ dz^* \end{bmatrix} = \begin{bmatrix} \frac{\partial x^*}{\partial x} & \frac{\partial x^*}{\partial y} & \frac{\partial x^*}{\partial z} \\ \frac{\partial y^*}{\partial x} & \frac{\partial y^*}{\partial y} & \frac{\partial y^*}{\partial z} \\ \frac{\partial z^*}{\partial x} & \frac{\partial z^*}{\partial y} & \frac{\partial z^*}{\partial z} \end{bmatrix} \begin{bmatrix} dx \\ dy \\ dz \end{bmatrix}, \quad (2.285)$$

or, in a more condensed notation as

$$d\mathbf{x}^* = \mathbf{F} d\mathbf{x}, \quad (2.286)$$

where the matrix $\mathbf{F}$, whose elements are given in (2.285), is known as the *deformation gradient* (not to be confused with the displacement gradient, to which it is related, as shown below).

The deformation gradient $\mathbf{F}$ contains within it all of the relevant information concerning the stretching and rotation of the various elements of the deformed rock. It does not serve directly as a measure of strain, however, because a suitable measure of finite strain should have the property of vanishing if, for example, the rock undergoes a rigid-body rotation. To construct a suitable finite strain tensor, we examine the change in *length* undergone by the element $d\mathbf{x}$:

$$|d\mathbf{x}|^2 = d\mathbf{x} \cdot d\mathbf{x} = d\mathbf{x}^T d\mathbf{x}, \quad (2.287)$$

$$|d\mathbf{x}^*|^2 = (d\mathbf{x}^*)^T d\mathbf{x}^* = (\mathbf{F} d\mathbf{x})^T (\mathbf{F} d\mathbf{x}) = d\mathbf{x}^T (\mathbf{F}^T \mathbf{F}) d\mathbf{x}, \quad (2.288)$$

$$|d\mathbf{x}|^2 - |d\mathbf{x}^*|^2 = d\mathbf{x}^T d\mathbf{x} - d\mathbf{x}^T (\mathbf{F}^T \mathbf{F}) d\mathbf{x} = d\mathbf{x}^T (\mathbf{I} - \mathbf{F}^T \mathbf{F}) d\mathbf{x} \equiv 2d\mathbf{x}^T \mathbf{E} d\mathbf{x}, \quad (2.289)$$

where $\mathbf{E}$, as defined in (2.289), is the *Lagrangian finite strain tensor*. We have defined it here so as to be consistent with the “compression = positive” sign convention; it therefore differs from the usual definition given in continuum mechanics treatments, such as Fung (1965) or Malvern (1969). It can be seen from (2.289) that the elongation (or shortening, as the case may be) of a linear element can be computed from $\mathbf{E}$. Note that if the deformation is a rigid-body rotation, then it will necessarily be true that $\mathbf{F}^T = \mathbf{F}^{-1}$, in which case $\mathbf{F}^T \mathbf{F} = \mathbf{F} \mathbf{F}^{-1} = \mathbf{I}$, and the Lagrangian strain vanishes. This can be verified (in two dimensions) by considering a rigid-body rotation, such as given by (2.199), and explicitly calculating $\mathbf{F}^T \mathbf{F}$. Although $\mathbf{F}$ will not necessarily be symmetric, it is easy to verify that $\mathbf{E}$ is symmetric.

Each normal component of $\mathbf{E}$ provides a measure of the strain of an element that initially lies along one of the coordinate axes. For example, consider an element that initially lies along the x -axis, that is, $d\mathbf{x} = (dx, 0, 0)$. From (2.287), we have

$$|d\mathbf{x}|^2 = \begin{bmatrix} dx & 0 & 0 \end{bmatrix} \begin{bmatrix} dx \\ 0 \\ 0 \end{bmatrix} = (dx)^2, \quad (2.290)$$

and from (2.289), we have

$$|d\mathbf{x}|^2 - |d\mathbf{x}^*|^2 = 2 \begin{bmatrix} dx & 0 & 0 \end{bmatrix} \begin{bmatrix} E_{xx} & E_{xy} & E_{xz} \\ E_{yx} & E_{yy} & E_{yz} \\ E_{zx} & E_{zy} & E_{zz} \end{bmatrix} \begin{bmatrix} dx \\ 0 \\ 0 \end{bmatrix} = 2E_{xx}(dx)^2, \quad (2.291)$$

which is to say

$$|d\mathbf{x}^*| = (1 - 2E_{xx})^{1/2} |d\mathbf{x}|. \quad (2.292)$$

The fractional shortening, as defined in (2.169), is therefore given by

$$\frac{|d\mathbf{x}| - |d\mathbf{x}^*|}{|d\mathbf{x}|} = 1 - (1 - 2E_{xx})^{1/2}. \quad (2.293)$$

If E_{xx} is small, the right-hand side of (2.293) reduces to E_{xx} , and we recover (2.181).

Another commonly used measure of the elongation is the *stretch ratio*, λ , defined by

$$\lambda \equiv \frac{|d\mathbf{x}^*|}{|d\mathbf{x}|} = (1 - 2E_{xx})^{1/2}. \quad (2.294)$$

The stretch ratio of an element initially lying along the y direction would be $(1 - 2E_{yy})^{1/2}$, etc. As there may also, in general, be a rotational component to the deformation, a line segment that initially lies in the x direction will not necessarily continue to lie in that direction after the deformation.

Now consider two infinitesimal elements that are initially at right angles to each other, for example, $\mathbf{dx}_1 = (dx, 0, 0)$ and $\mathbf{dx}_2 = (0, dy, 0)$. These elements can be identified with PQ and PR of Fig. 2.12a. We now take the inner product of these two elements after deformation, as in the calculations given by (2.214)–(2.221):

$$\begin{aligned} |\mathbf{dx}_1^*| |\mathbf{dx}_2^*| \cos \angle R^* P^* Q^* &= (\mathbf{dx}_1^*) \cdot (\mathbf{dx}_2^*) = (\mathbf{dx}_1^*)^T (\mathbf{dx}_2^*) \\ &= (\mathbf{Fdx}_1)^T (\mathbf{Fdx}_2) = (\mathbf{dx}_1)^T \mathbf{F}^T \mathbf{F} (\mathbf{dx}_2) = (\mathbf{dx}_1)^T (\mathbf{I} - 2\mathbf{E}) (\mathbf{dx}_2) = [dx \ 0 \ 0] \\ &\quad \times \begin{bmatrix} 1 - 2E_{xx} & -2E_{xy} & -2E_{xz} \\ -2E_{yx} & 1 - 2E_{yy} & -2E_{yz} \\ -2E_{zx} & -2E_{zy} & 1 - 2E_{zz} \end{bmatrix} \begin{bmatrix} 0 \\ dy \\ 0 \end{bmatrix} = -2E_{xy} dx dy. \end{aligned} \quad (2.295)$$

The lengths of the deformed elements $|\mathbf{dx}_1^*|$ and $|\mathbf{dx}_2^*|$ are given by (2.292), so

$$\begin{aligned} \cos \angle R^* P^* Q^* &= \frac{-2E_{xy} dx dy}{(1 - 2E_{xx})^{1/2} dx (1 - 2E_{yy})^{1/2} dy} \\ &= \frac{-2E_{xy}}{(1 - 2E_{xx})^{1/2} (1 - 2E_{yy})^{1/2}}. \end{aligned} \quad (2.296)$$

Hence, the change in the angle formed by two initially perpendicular elements can be found in terms of the components of the finite strain tensor, although the relationship is not as simple as in the infinitesimal case. If all of the strain components are small, then, following along the lines of (2.220)–(2.221), it can be shown that the increase in the angle is given by $2E_{xy}$, and we recover (2.221).

The relationship between the finite strain tensor and the infinitesimal strain tensor can be investigated more thoroughly by recalling that, for example, $x^* = x - u$, in which case

$$\frac{\partial x^*}{\partial x} = 1 - \frac{\partial u}{\partial x}; \quad \frac{\partial x^*}{\partial y} = -\frac{\partial u}{\partial y}; \quad \frac{\partial x^*}{\partial z} = -\frac{\partial u}{\partial z}, \quad (2.297)$$

and similarly for y^* and z^* . Therefore, $\mathbf{F} = \mathbf{I} - \nabla \mathbf{u}$, and so

$$\begin{aligned} 2\mathbf{E} &= \mathbf{I} - \mathbf{F}^T \mathbf{F} = \mathbf{I} - (\mathbf{I} - \nabla \mathbf{u})^T (\mathbf{I} - \nabla \mathbf{u}) = \mathbf{I} - [\mathbf{I} - (\nabla \mathbf{u})^T] [\mathbf{I} - \nabla \mathbf{u}] \\ &= \mathbf{I} - [\mathbf{I} - (\nabla \mathbf{u})^T - \nabla \mathbf{u} + (\nabla \mathbf{u})^T \nabla \mathbf{u}] = (\nabla \mathbf{u})^T + \nabla \mathbf{u} - (\nabla \mathbf{u})^T \nabla \mathbf{u}, \\ \text{that is, } \mathbf{E} &= \frac{1}{2} [\nabla \mathbf{u} + (\nabla \mathbf{u})^T - (\nabla \mathbf{u})^T \nabla \mathbf{u}]. \end{aligned} \quad (2.298)$$

Comparison of (2.298) with (2.223) shows that in addition to terms that are *linear* in the components of the displacement gradient, the finite strain tensor also contains terms that are *quadratic* in the components of the displacement gradient. If all of the components of $\nabla \mathbf{u}$ are small, the quadratic terms can be neglected, and $\mathbf{E}$ reduces to the infinitesimal strain tensor, $\mathbf{e}$. The individual components of $\mathbf{E}$, as defined by (2.298), are

$$E_{xx} = \frac{\partial u}{\partial x} - \frac{1}{2} \left[\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial x} \right)^2 + \left(\frac{\partial w}{\partial x} \right)^2 \right], \quad (2.299)$$

$$E_{yy} = \frac{\partial v}{\partial y} - \frac{1}{2} \left[\left(\frac{\partial u}{\partial y} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2 + \left(\frac{\partial w}{\partial y} \right)^2 \right], \quad (2.300)$$

$$E_{zz} = \frac{\partial w}{\partial z} - \frac{1}{2} \left[\left(\frac{\partial u}{\partial z} \right)^2 + \left(\frac{\partial v}{\partial z} \right)^2 + \left(\frac{\partial w}{\partial z} \right)^2 \right], \quad (2.301)$$

$$E_{xy} = E_{yx} = \frac{1}{2} \left[\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} - \left(\frac{\partial u}{\partial x} \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \frac{\partial v}{\partial y} + \frac{\partial w}{\partial x} \frac{\partial w}{\partial y} \right) \right], \quad (2.302)$$

$$E_{xz} = E_{zx} = \frac{1}{2} \left[\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} - \left(\frac{\partial u}{\partial x} \frac{\partial u}{\partial z} + \frac{\partial v}{\partial x} \frac{\partial v}{\partial z} + \frac{\partial w}{\partial x} \frac{\partial w}{\partial z} \right) \right], \quad (2.303)$$

$$E_{yz} = E_{zy} = \frac{1}{2} \left[\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} - \left(\frac{\partial u}{\partial y} \frac{\partial u}{\partial z} + \frac{\partial v}{\partial y} \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \frac{\partial w}{\partial z} \right) \right]. \quad (2.304)$$